MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

STATEMENT-1 : Average power consumed in an any ac circuit is always equal to average power consumed by resistors in the circuit.

STATEMENT-2 : Average power consumed in an any ac circuit by capacitor and inductor is zero.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Q.2

A plate of mass (M) is placed on a horizontal frictionless surface and a body of mass (m) is placed on this plate. The coefficient of dynamic friction between this body and the plate is μ. If a force $3\mu\mathrm{mg}$ is applied to the body of mass (m) along the horizontal, the acceleration of the plate will be

(A) $\frac{\mu \mathrm{m}}{\mathrm{M}} \mathrm{g}$

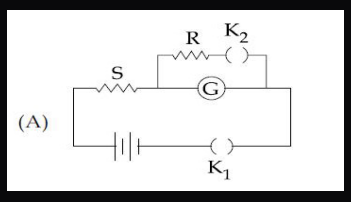
(B) $\frac{\mu \mathrm{mg}}{\mathrm{M}+\mathrm{m}}$

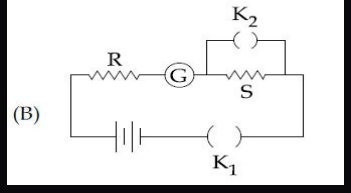
(C) $\frac{3\mu \mathrm{mg}}{\mathrm{M}}$

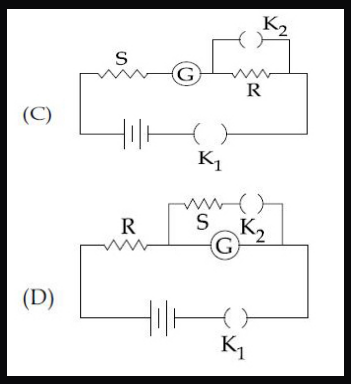
(D) $\frac{2\mu \mathrm{mg}}{\mathrm{M}+\mathrm{m}}$

Q.3

In the circuit diagrams (A, B, C and D) shown below, R is high resistance and S is a resistance of the order of galvanometer resistance G. The correct circuit, corresponding to the half deflection method for finding the resistance and figure of merit of the galvanometer, is the circuit labelled as:







(A) Circuit D with G $= \frac{RS}{R-S}$

(B) Circuit A with G $= \frac{RS}{(R-S)}$

(C) Circuit C with G = S

(D) Circuit B with G = S

Q.4

A force is given by $F = at + bt^2$, where, t is the time. What are the dimensions of a and b?

(A) $[MLT^{-4}]$ and $[MLT]$

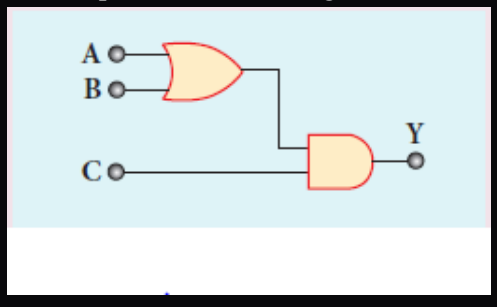
(B) $[MLT^{-1}]$ and $[MLT^0]$

(C) $[MLT^{-3}]$ and $[MLT^{-4}]$

(D) $[MLT^{-3}]$ and $[MLT^0]$

Q.5

The output of the following circuit is 1 when the input ABC is



(A) 101

(B) 100

(C) 110

(D) 010

Q.6

A stone of relative density K is released from rest on the surface of a lake. If viscous effects are ignored, the stone sinks in water with an acceleration of,

(A) $g(1-K)$

(B) $g(1+K)$

(C) $g\left(1-\frac{1}{K}\right)$

(D) $g\left(1+\frac{1}{K}\right)$

Q.7

The angular velocity of the earth with which it has to rotate so that acceleration due to gravity on 60° latitude becomes zero is (Radius of the earth= 6400 km. At the poles $g=10 \mathrm{~m~s}^{-2}$)

(A) $2.5 \times 10^{-3} \mathrm{~rad~s}^{-1}$

(B) $5.0 \times 10^{-1} \mathrm{~rad~s}^{-1}$

(C) $100 \mathrm{~rad~s}^{-1}$

(D) $7.8 \times 10^{-2} \mathrm{~rad~s}^{-1}$

Q.8

A hollow cylinder has a charge q placed at its centre. If ϕ is the electric flux (in Vm) associated with the curved surface B, the flux linked with the plane surface A is



(A) $\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

(B) $\frac{q}{2 \varepsilon_0}$

(C) $\frac{\phi}{3}$

(D) $\frac{q}{\varepsilon_0}-\phi$

Q.9

A hoop and a solid cylinder of same mass and radius are made of a permanent magnetic material with their respective axes. But the magnetic moment of hoop is twice of solid cylinder. They are placed in a uniform magnetic field in such a manner that their magnetic moments make a small angle with the field. If the oscillation periods of hoop and cylinder are T_h and T_c respectively, then:

(A) $T_h=2T_c$

(B) $T_h=T_c$

(C) $T_h=0.5T_c$

(D) $T_h=1.5T_c$

Q.10

Identify the correct statement(s) from among the following
(i) The constancy of the binding energy per nucleon in the range $30 < A < 170$ is a consequence of the fact that the nuclear force is short ranged.
(ii) The nuclear force does not depend on the charge of nucleons.
(iii) The nuclear force is repulsive when distance between two nucleons is less than $0.8\,\mathrm{fm}$

(A) (i) only

(B) (ii), (iii)

(C) (i), (ii)

(D) (i), (ii), (iii)

Q.11

Magnetic moment of bar magnet is *M*. The work done to change the alignment from parallel to perpendicular to the direction of the magnetic field B will be

(A) Zero

(B) $\frac{1}{2}MB$

(C) $2MB$

(D) MB

Q.12

A uniform force of $\left(3\hat{\mathrm{i}}+\hat{\mathrm{j}}\right)$ newton acts on a particle of mass 2 kg . Hence the particle is displaced from position $\left(2\hat{\mathrm{i}}+\widehat{\mathrm{k}}\right)$ metre to position $\left(4\hat{\mathrm{i}}+3\hat{\mathrm{j}}-\widehat{\mathrm{k}}\right)$ metre. The work done by the force on the particle is

(A) 9 J

(B) 6 J

(C) 13 J

(D) 15 J

Q.13

Which of the following statements about the specific heat of a body is/are correct?
1. It depends upon the mass and shape of the body.
2. It is independent of the mass and shape of the body.
3. It depends only upon the temperature of the body.
Select the correct answer using the codes given below:

(A) 2 and 3

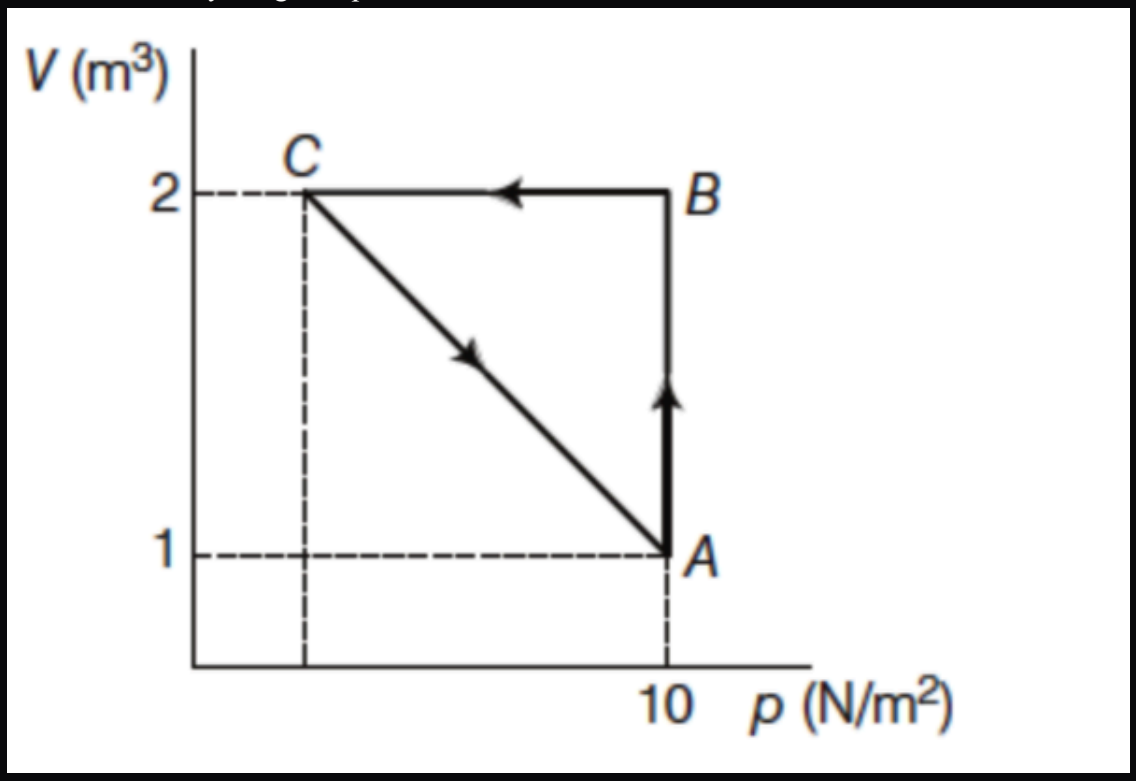
(B) 1 only

(C) 1 and 3

(D) 2 only

Q.14

An ideal gas is taken through a cycle $A \to B \to C \to A$, as shown. If the net heat supplied to the gas in the cycle is $5\ \text{J}$, then workdone by the gas in process $C \to A$ is



(A) $-5\ \text{J}$

(B) $-10\ \text{J}$

(C) $-15\ \text{J}$

(D) $-20\ \text{J}$

Q.15

The ratio of the most probable velocity (c_m) and rms velocity (c_{rms}) of the molecules of a given gas at a given temperature is

(A) $\sqrt{\frac{3}{2}}$

(B) $\sqrt{\frac{2}{3}}$

(C) $\sqrt{\frac{3}{4}}$

(D) $\frac{2}{\sqrt{3}}$

Q.16

Assertion (A) : The speed of a body in uniform circular motion is constant.
Reason (B) : In uniform circular motion, the acceleration of the body is constant.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

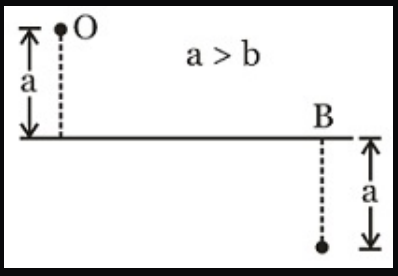
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (B) is true

Q.17

A luminous point is placed at O, whose image is formed at I as shown in diagram. Line AB is optic axis and its length is given. Which of the following statements are correct?



(A) If the lens used to obtain the image, the lens must be a converging lens and position of its principal focus can be found out

(B) If the mirror is used to obtain the image, the mirror must be a convex mirror whose pole will be at point of intersection of line OI with the optic axis

(C) Position of principal focus of the mirror cannot be found out

(D) I is a real image of virtual object

Q.18

A body of mass 100 g moves at the speed of 36 km h^{-1}. The de Broglie wavelength related to it is of the order _____m $\left(h = 6.626 \times 10^{-34}\ \text{Js}\right)$

(A) 10^{-14}

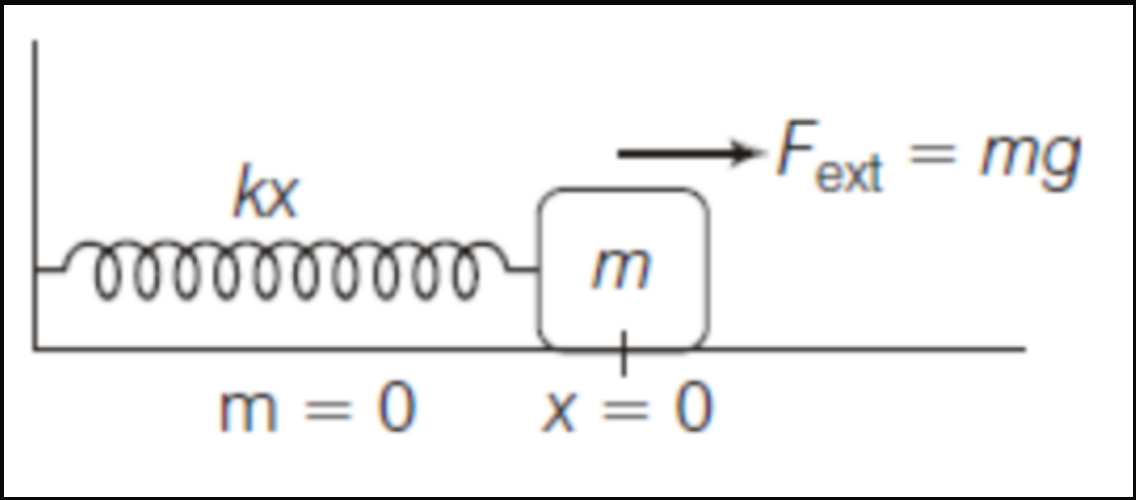
(B) 10^{-24}

(C) 10^{-34}

(D) 10^{-44}

Q.19

A force starts acting on a spring block system. At a stretch when $x = \frac{mg}{k}$, workdone by external force and spring force are related as



(A) $W_{F_{ext}} + W_{F_s} = 0$

(B) $W_{F_{ext}} = \frac{1}{2}kx^2$

(C) $W_{F_s} + W_{F_{ext}} > 0$

(D) None of these

Q.20

A plane electromagnetic wave is propagating along the positive z direction. If the electric field component of this wave is in the direction $(\hat{i}+\hat{j})$, then which of the following is the direction of the magnetic field component?

(A) $(-\hat{i}+\hat{j})$

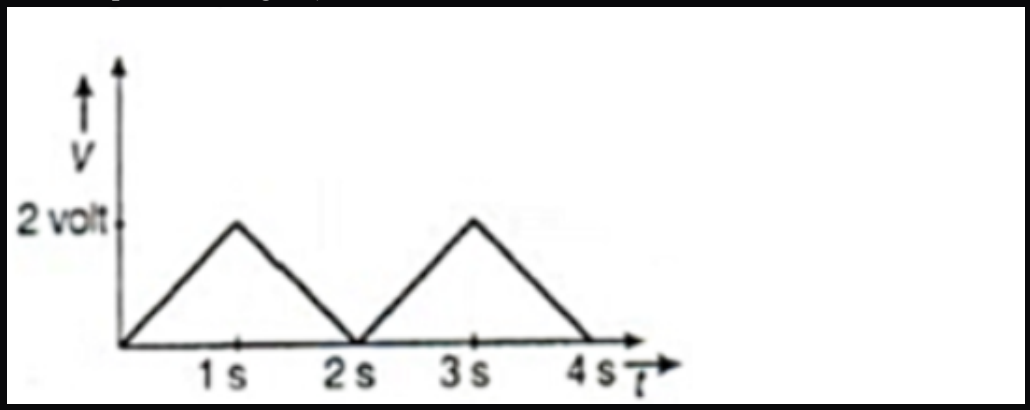
(B) $(\hat{i}-\hat{j})$

(C) $(-\hat{i}-\hat{j})$

(D) $(\hat{i}+\hat{k})$

Q.21

If the rms value of the voltage of the waveform shown below is $\sqrt{\frac{p}{q}}\ \mathrm{V}$, then what is the value of $(p+q)$?(p and q are the smallest positive integers.)



Q.22

The relative density of the material of a body is the ratio of its weight in air and the loss of its weight in water. By using a spring balance, the weight of the body in the air is measured to be 5.00 ± 0.05 N. The weight of the body in water is measured to be 4.00 ± 0.05 N. Then the maximum possible percentage error in relative density is (mark answer in percentage)

Q.23

A boy is pushing a ring of mass $2\ \mathrm{kg}$ and radius $0.5\ \mathrm{m}$ with a stick as shown in the figure. The stick applies a force of $2\ \mathrm{N}$ on the ring and rolls it without slipping with an acceleration of $0.3\ \mathrm{m\ s^{-2}}$. The coefficient of friction between the ground and the ring is large enough so that rolling always occurs. The coefficient of friction between the stick and the ring is $P/10$. The value of P is

Q.24

All the edges of a block with parallel faces are unequal. Its largest edge is twice its shortest edge. The ratio of the maximum to the minimum resistance between parallel faces is

Q.25

A golfer standing on the level ground hits a ball with a velocity of $52\ \text{m s}^{-1}$ at an angle θ above the horizontal. If $\tan(\theta) = \frac{5}{12}$, then find the time for which the ball is at least $15\ \text{m}$ above the ground (take $g = 10\ \text{m s}^{-1}$).

Q.26

Heptane and octane form an ideal solution. At $373\ \text{K}$, the vapour pressures of the two liquid components are $105.2\ \text{kPa}$ and $46.8\ \text{kPa}$, respectively. What will be the vapour pressure of a mixture of $26.0\ \text{g}$ of heptane and $35\ \text{g}$ of octane?

(A) $7.308\ \text{kPa}$

(B) $73.08\ \text{kPa}$

(C) $730.8\ \text{kPa}$

(D) $7308\ \text{kPa}$

Q.27

Which of the characteristics is not common between $[\text{Cu(en)}_2]^{2+}$ and $[\text{Ni(dmg)}_2]$?

(A) Geometry of complexes

(B) Hybridisation of central metal cation

(C) Magnetic behaviour

(D) Number of stereoisomers

Q.28

The rate of $\text{S}_\text{N}1$ reaction is fastest in the hydrolysis of which of the following halides?

(A) $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$

(B) CH_3Br

(C) $(\text{CH}_3)_2\,\text{CHBr}$

(D) $(\text{CH}_3)_3\,\text{CBr}$

Q.29

Considering the elements B, C, N, F and Si, the correct order of their non-metallic character is

(A) B > C > Si > N > F

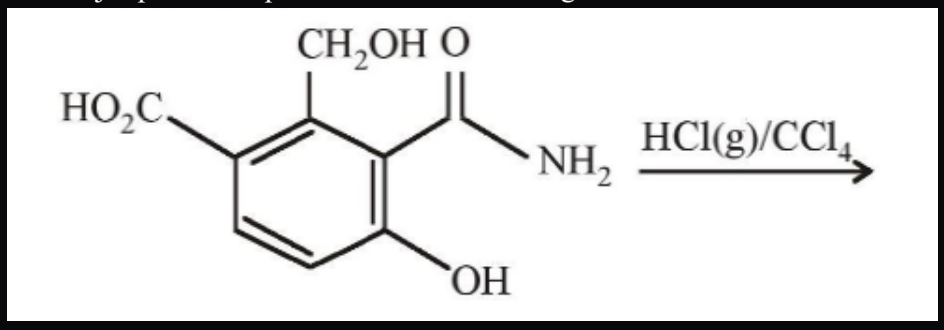
(B) Si > C > B > N > F

(C) F > N > C > B > Si

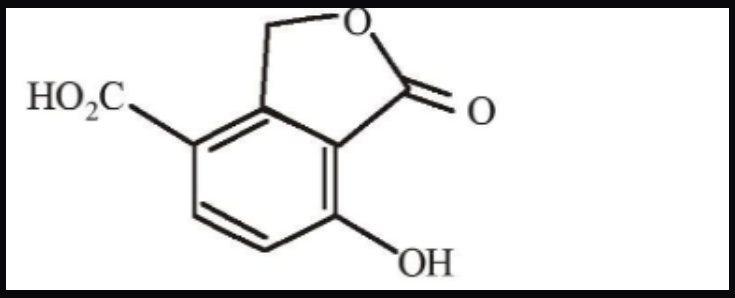
(D) F > N > C > Si > B

Q.30

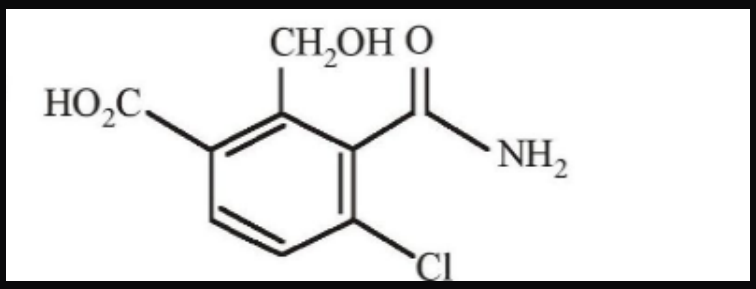
The major product expected from the following reaction is :



(A)



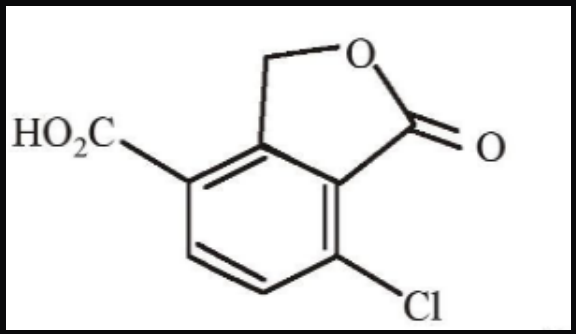
(B)



(C)



(D)



Q.31

An indicator has a pK_a of 7.5. What is the ratio of the concentration of the base form to the concentration of the acid form ($[A^-]/[HA]$) in a buffer solution in which the pH is 6.5 ?

(A) 10/1

(B) 1/1

(C) 1/10

(D) 6.5/7.5

Q.32

One mole of monoatomic ideal gas expands adiabatically at initial temperature T against a constant external pressure of 1 atm from 1 L to 2 L. Find out the final temperature ($\mathrm{R} = 0.0821\ \mathrm{L\ atm\ K^{-1}\ mol^{-1}}$).

(A) T

(B) $\frac{\mathrm{T}}{(2)^{\frac{5}{3}-1}}$

(C) $\mathrm{T} - \frac{2}{3\times0.082}$

(D) $\mathrm{T} + \frac{2}{3\times0.082}$

Q.33

There are four conformational isomers of n-butane:
Eclipsed, Fully eclipsed, Gauche, Staggered
Arrange these isomers of n-butane in increasing order of energy.

(A) Gauche < Eclipsed < Staggered < Fully eclipsed

(B) Staggered < Gauche < Eclipsed < Fully eclipsed

(C) Fully eclipsed < Eclipsed < Gauche < Staggered

(D) Gauche > Eclipsed < Staggered < Fully eclipsed

Q.34

A compound which is a strong oxidizing agent and has orange coloured crystal. It is used in the preparation of azo compounds. Identify the compound:

(A) Hydrogen peroxide

(B) Potassium permanganate

(C) Sodium chromate

(D) Potassium dichromate

Q.35

Molar ionic conductivities of two-bivalent electrolytes x^{2+} and y^{2-} are 57 and 73, respectively. The molar conductivity of the solution formed by them will be

(A) $130\ \mathrm{S\ cm^2\ \ mol^{-1}}$

(B) $65\ \text{S cm}^2\ \text{mol}^{-1}$

(C) $260\ \text{S cm}^2\ \text{mol}^{-1}$

(D) $187\ \text{S cm}^2\ \text{mol}^{-1}$

Q.36

Consider the molecules CH_4, NH_3 and H_2O. Which of the given statements is false?

(A) The H - C - H bond angle in CH_4, the H - N - H bond angle in NH_3, and the H - O - H bond angle in H_2O are all greater than 90°.

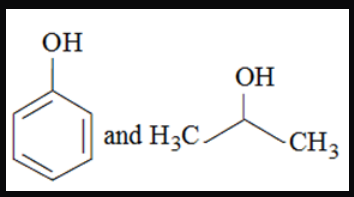
(B) The H - O - H bond angle in H_2O is larger than the H - C - H bond angle in CH_4.

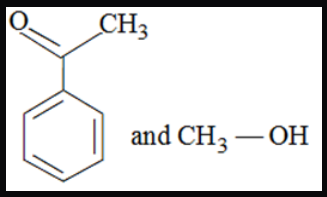
(C) The H - O - H bond angle in H_2O is smaller than the H - N - H bond angle in NH_3.

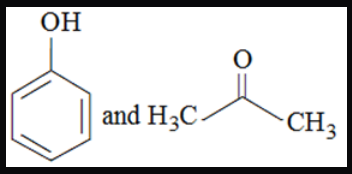
(D) The H - C - H bond angle in CH_4 is larger than the H - N - H bond angle in NH_3.

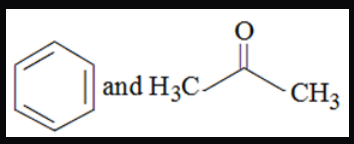
Q.37

The product formed in the reaction of Cumene with O_2 followed by treatment with dil.H_2SO_4 and heat are

(A) 

(B) 

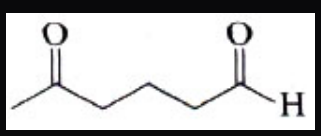
(C) 

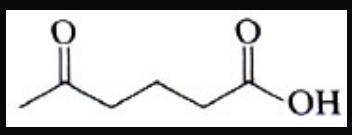
(D) 

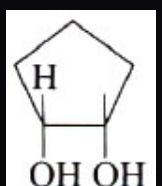
Q.38

What is the major product of the reaction :

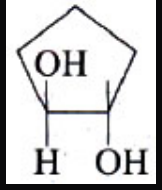
Methyl cyclopentene $+ KMnO_4/H^+/\text{Cold} \rightarrow$?

(A) 

(B) 

(C) 

(D)



Q.39

The number of HCl molecules present in $10\ \text{mL}$ of $0.1\ \text{N}$ HCl solution is:
(Hint: Normality $=$ n $-$ factor $\times$ molarity)

(A) $6 \cdot 022 \times 10^{23}$

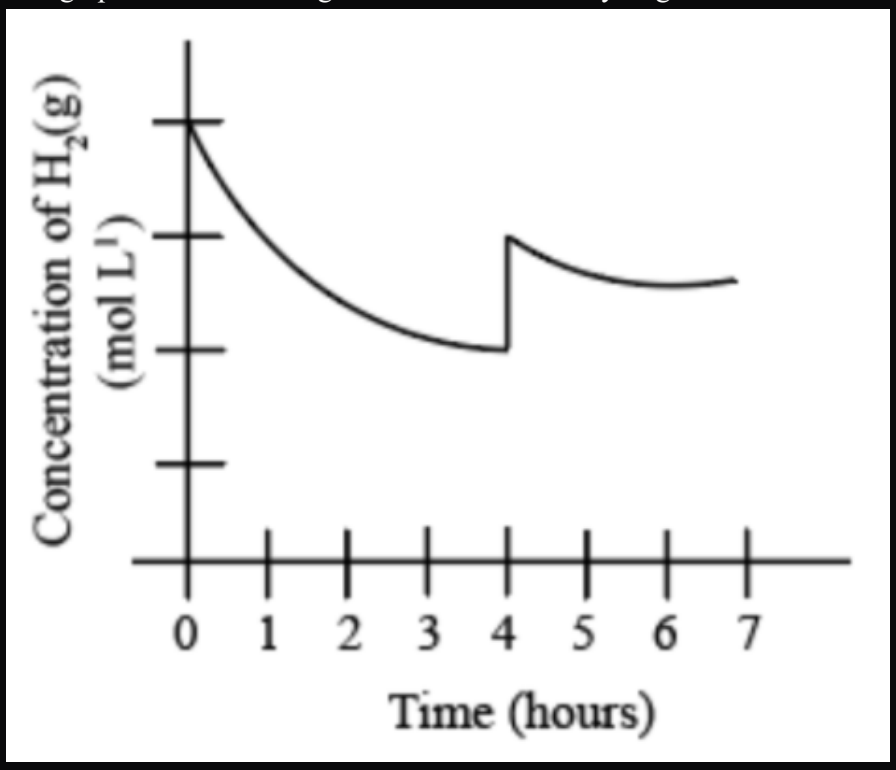
(B) $6 \cdot 022 \times 10^{22}$

(C) $6 \cdot 022 \times 10^{21}$

(D) $6 \cdot 022 \times 10^{20}$

Q.40

The reaction between phosphorus (P_4) and hydrogen (H_2) can result in the formation of phosphine (PH_3) as shown:
$\text{P}_4(\text{ s}) + 6\text{H}_2(\text{ g}) \rightleftharpoons 4\text{PH}_3(\text{ g})$
The graph shows the change in concentration of hydrogen for this reaction in which the system was disturbed after four hours.



Which of the following could explain the change in hydrogen concentration at time $= 4$ hours?

(A) The volume on the reaction vessel was decreased

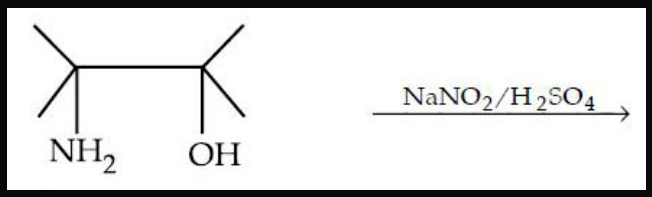
(B) A catalyst was added

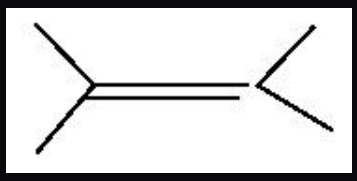
(C) The pressure on the reaction mixture has no effect.

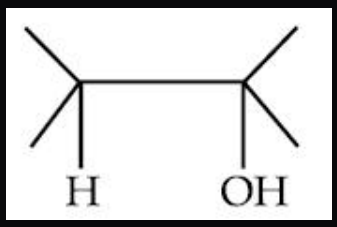
(D) More phosphorus was added

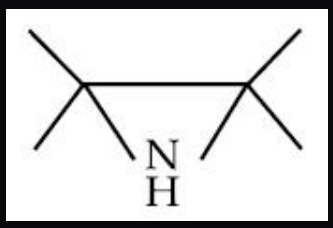
Q.41

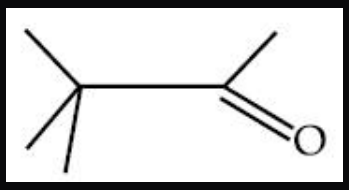
The major product of the reaction is



(A) 

(B) 

(C) 

(D) 

Q.42

Assertion : Si – Si bonds are much weaker than Si – O bonds.
Reason : Silicon forms double bonds with itself.

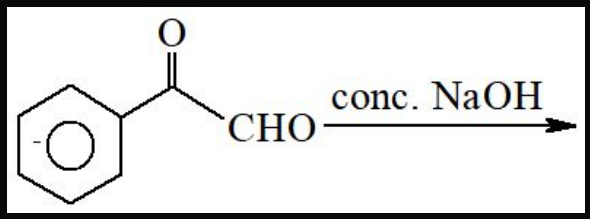
(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

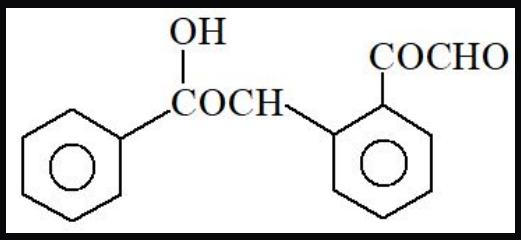
(D) Assertion and reason are false

Q.43

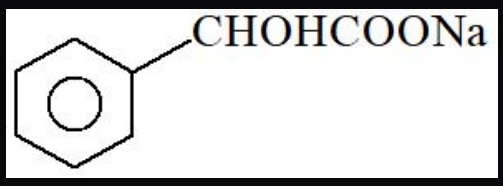


Product should be

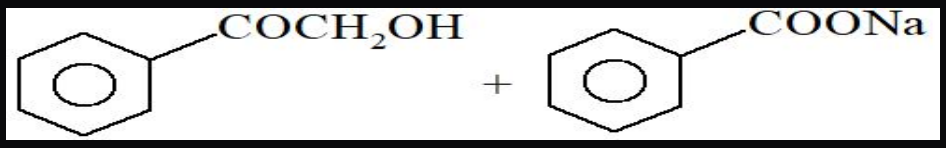
(A)



(B)



(C)



(D)



Q.44

Match list I with list II and select this correct answer using the codes given below the lists.

List-I	**List-II**
(p) Nucleic acids	1. DNA
(q) Uracil	2. Hormones
(r) Thymine	3. Polynucleotides
(s) Double-helix structure	4. RNA

(A) (p)-4, (q)-3, (r)-1, (s)-1

(B) (p)-3, (q)-4, (r)-1, (s)-1

(C) (p)-3, (q)-1, (r)-4, (s)-1

(D) (p)-3, (q)-1, (r)-1, (s)-4

Q.45

The screening effect of inner electrons will lead to

a. Decrease in the ionization potential.

b. Increase in the ionization potential.

c. Decrease in electron affinity.

d. Increase in electron affinity.

(A) Both (a) and (c).

(B) Both (a) and (d).

(C) Both (b) and (c).

(D) Both (b) and (d).

Q.46

29.5 mg of organic compound containing nitrogen was digested according to Kjeldahl's method and the evolved ammonia was absorbed in 20mL of 0.1 M HCl solution. The excess of the acid required 15mL of 0.1M NaOH solution for complete neutralisation. The percentage of nitrogen in the compound is (mark answer to nearest integer)

Q.47

For a dilute solution containing 2.5 g of a non-volatile non–electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is $2^{\circ}C$. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (in mm of Hg) of the solution is (take $K_b = 0.76$ Kkg mol^{-1})

Q.48

For the first order reaction
$C_2H_4O_{(g)} \rightarrow CH_{4(g)} + CO_{(g)}$,
the initial pressure of $C_2H_4O_{(g)}$ is 80 torr and total pressure at the end of 20 minutes is 120 torr. The time needed for 75% decomposition of C_2H_4 would be:

Q.49

The energy of the election in the second and third Bohr's orbits of the hydrogen atom is - 5.42 x 10^{-12} erg and - 2.41 x 10^{-12} erg respectively. Calculate the wavelength of the emitted light when the electron drops from the third to the second orbit, in nanometer.

Q.50

Number of moles of MnO_4^- required to oxidise five moles of ferrous oxalate completely in acidic medium will be ..... mol.

Q.51

Consider the letters of the word 'INCURSION'. Number of arrangements of all letters in which vowels are in alphabetical order, is

(A) $\frac{9!}{2!^2\ 4!}$

(B) $\frac{9!}{2!\ 4!}$

(C) ${}^9C_4$

(D) ${}^9P_5$

Q.52

If $\lim\limits_{x \to 0}\left(x^{-3}\ \sin\ 3x + ax^{-2} + b\right)$ exists and is equal to 0, then

(A) $a = -3$ and $b = 9/2$

(B) $a = 3$ and $b = 9/2$

(C) $a = -3$ and $b = -9/2$

(D) $a = 3$ and $b = -9/2$

Q.53

The value of $I = \int_{-1}^{1}[x \sin(\pi x)]dx$ is (where $[\cdot]$ denotes the greatest integer function)

(A) π

(B) 2π

(C) 0

(D) $-\pi$

Q.54

A biased coin is tossed repeatedly until a tail appears for the first time. Heads is 3 times as likely to appear as tails. Let x be the number of tosses required. Assume that all the trials of tossing a biased coin are independent. Then, the conditional probability that $x \geq 6$, given that $x > 3$, is equal to

(A) $\frac{9}{16}$

(B) $\frac{1}{16}$

(C) $\frac{1}{2}$

(D) $\frac{3}{5}$

Q.55

If PQR is an equilateral triangle inscribed in the auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$ and $P'Q'R'$ is the corresponding triangle inscribed within the ellipse, then centroid of the triangle $P'Q'R'$ lies at

(A) centre of ellipse

(B) focus of ellipse

(C) between focus and centre on major axis

(D) none of these

Q.56

Let S be the relation on N defined by $S = \left\{(x, y) : \ \frac{y}{2} \leq x \leq 2y\right\}$. Then, S is

(A) reflexive only

(B) transitive only

(C) symmetric only

(D) reflexive and symmetric

Q.57

Let ΔOAB be an equilateral triangle with side length unity (O being the origin). Also, M and N are the points of

trisection of AB, M being closer to A and N being closer to B. Position vectors of A, B, M and N are $\vec{a}, \vec{b}, \vec{m}$ and $\vec{n}$ respectively, then the value of $\vec{m} \cdot \vec{n}$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{2}{3}$

(C) $\frac{13}{18}$

(D) $\frac{4}{9}$

Q.58

Consider $f : R^+ \to R$ such that $f(3) = 1$ for $a \in R^+$ and $f\left(x\right) \cdot f\left(y\right) + f(\frac{3}{x}) f\left(\frac{3}{y}\right) = 2f\left(xy\right) \forall x, y \in R^+$ then $f(97)$ can be

(A) 1

(B) -1

(C) 2

(D) 97

Q.59

Let $A,\ B,\ C$ and D are any 4 points in space. If $\left|\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}\right| = \lambda\ (\Delta ABC)$, where ΔABC is the area of triangle ABC, then the value of λ is equal to

(A) 2

(B) $\frac{1}{2}$

(C) 4

(D) $\frac{1}{4}$

Q.60

If $P = \{1,2,3,4,5\}$ and $Q = \{a, b, c\}$, then the number of onto functions from P to Q is

(A) 150

(B) 144

(C) 147

(D) 154

Q.61

Consider the function $f\left(x\right) = \int\limits_0^x \left(5\ In\left(1 + t^2\right) - 10t\ \tan^{-1} t + 16 \sin t\right) dt.\ f\left(x\right)$ is

(A) Negative for all $x \in (0, 1)$

(B) Increasing for all $x \in (0, 1)$

(C) Non-monotonic for all $x \in (0, 1)$

(D) None of these

Q.62

If $a = \underbrace{111\ldots\ldots1}_{55 \text{ times}}, b = 1 + 10 + 10^2 + 10^3 + 10^4$ and $c = 1 + 10^5 + 10^{10} + \ldots. + 10^{50}$, then

(A) $b, \frac{a}{2}, c$ are in arithmetic progression

(B) $b, \sqrt{a}, c$ are in geometric progression

(C) a, b, c are in geometric progression

(D) $a, \sqrt{b}, c$ are in arithmetic progression

Q.63

A box contains 9 slips bearing numbers $-3,\ -2,\ -1, 0,\ 1, 2,\ 3, 4$ and 5. An experiment consists of drawing a slip from this box and replacing it back in the box after noting the number. This experiment is repeated 9 times. These 9 numbers are now chosen as elements of a 3×3 matrix, then the probability that the matrix is skew symmetric is

(A) $\frac{1}{9^6}$

(B) $\frac{343}{9^9}$

(C) $\frac{1}{9^9}$

(D) $\frac{1}{9^7}$

Q.64

Statement-1: The direction cosines of a pair of lines satisfy $\lambda(l + m) = n$ and $mn + nl + lm = 0$, where $l,\ m,\ n$ are direction cosines. The two lines are perpendicular if $\lambda = 2$.
Statement-2: Lines with direction cosines $l_1,\ m_1,\ n_1$ and $l_2,\ m_2,\ n_2$ are perpendicular if $l_1l_2 + m_1m_2 + n_1n_2 = 0$.

(A) If both the statements are true and statement-2 is the correct explanation of statement-1.

(B) If both the statements are true but statement-2 is not the correct explanation of statement-1.

(C) If statement-1 is true and statement-2 is false.

(D) If statement-1 is false and statement-2 is true.

Q.65

If $f(x)$ be a differentiable function such that $\frac{d^n}{dx^n} f\left(x\right)$ exists $\forall x \in R,\ n \in N$ and $f'(\alpha) = f'(\beta) = 0,\ f''(\alpha) \cdot f''(\beta) < 0$, then which of the following statements can be true

(A) points $(\alpha, f(\alpha)), (\beta, f(\beta))$ are local maxima

(B) points $(\alpha, f(\alpha)), (\beta, f(\beta))$ are local minima

(C) $(\alpha, f(\alpha)), (\beta, f(\beta))$ are points of inflection

(D) none of these

Q.66

The term independent of x in the expansion of $\left[\left(t^{-1} - 1\right)x + \left(t^{-1} + 1\right)^{-1} x^{-1}\right]^8$ is :

(A) $56\left(\frac{1-t}{1+t}\right)^3$

(B) $56\left(\frac{1+t}{1-t}\right)^3$

(C) $70\left(\frac{1-t}{1+t}\right)^4$

(D) $70\left(\frac{1+t}{1-t}\right)^4$

Q.67

Find the area bounded by the curve $y = xe^{-x^2}$, the x-axis and the line $x = c$ where $y(c)$ is maximum.

(A) $\frac{1}{2}\left(1-\frac{1}{\sqrt{e}}\right)$ sq. units

(B) $\frac{1}{4}\left(1-e^{1/4}\right)$ sq. units

(C) $\frac{1}{3}\left(1-e^{1/3}\right)$ sq. units

(D) $\left(1-e^{1/2}\right)$ sq. units

Q.68

If a curve $y = f(x)$ passing through $(1,\ 2)$ satisfies the differential equation $y(1+xy)dx - xdy = 0$, then which of the following option is correct?

(A) $f(x) = \frac{2x}{2-x^2}$

(B) $f(x) = \frac{x+1}{x^2+1}$

(C) $f(x) = \frac{x-1}{4-x^2}$

(D) $f(x) = \frac{4x}{1-2x^2}$

Q.69

If $\left|\frac{z+i}{z-i}\right| = \sqrt{3}$, then radius of the circle is

(A) $\frac{2}{\sqrt{21}}$

(B) $\frac{1}{\sqrt{21}}$

(C) $\sqrt{3}$

(D) $\sqrt{21}$

Q.70

Let V_1 = variance of $\{13,16,19.\ldots\ldots\ldots 103\}$ and V_2 = variance of $\{20,26,32.\ldots\ldots\ldots 200\}$. Then $\mathrm{V}_1 : \mathrm{V}_2$ is

(A) $1:2$

(B) $1:1$

(C) $4:9$

(D) $1:4$

Q.71

If M and m are the maximum and minimum values of $\frac{y}{x}$ for pair of real numbers (x, y) which satisfy the equation $(x-3)^2+(y-3)^2=6$, then the value of $\frac{1}{\mathrm{M}}+\frac{1}{m}$ is

Q.72

Number of ordered pairs (z, ω) of the complex numbers z and ω, satisfying the system of equations, $z^3+(\bar{\omega})^7=0$ and $z^5 \cdot \omega^{11}=1$ is

Q.73

Let the lengths of the altitudes from the vertices $A(-1,1), B(5,2), C(3,-1)$ of ΔABC are p_1, p_2, p_3 units respectively, then the value of $\frac{\left(\frac{1}{p_1}\right)^2+\left(\frac{1}{p_3}\right)^2}{\left(\frac{1}{p_2}\right)^2}$ is equal to a/b, then what is the value of a-b ? (a and b are coprime positive integers)

Q.74

The number of ways in which 8 non-identical apples can be distributed among 3 boys such that every boy should get atleast 1 apple and atmost 4 apples is $k \cdot {}^7P_3$, where k has the value equal to

Q.75

Let the sets $A=\{2,4,6,8 \ldots\}$ and $B=\{3,6,9,12 \ldots\}$ such that $n(A)=200$ and $n(B)=250$. If $n(A \cup B)=k$, then k is equal to

Answers & Solutions

Q.1 Answer:

—

Solution:

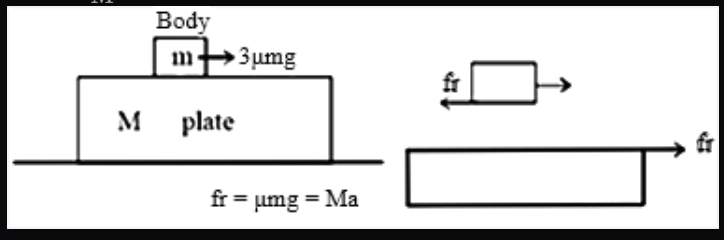
For ideal inductors and capacitors there will be no dissipation of enerw. They first take energies and then release them again for an AC circuit.
But in resistors there is dissipation of enerw which equals i^2R
Average power consumed in an ac circuit is equal to average power consumed by resistors in the circuit.

Q.2 Answer:

$\frac{\mu \mathrm{m}}{\mathrm{M}} \mathrm{g}$

Solution:

$\mathrm{a}=\frac{\mu \mathrm{mg}}{\mathrm{M}}$



First we will assume that both block and plate will be moving together. If the common acceleration comes out to

be greater than the maximum acceleration plate can have then we will treat block and plate seperately.

$\mu \mathrm{N} = \mu \mathrm{mg} \quad (\max \text{ friction between block and plate})$

$\max$ acceleration of plate is $\frac{\mu \mathrm{mg}}{\mathrm{M}}$

But if they are moving together then,

$\mathrm{a} = \frac{\mathrm{F}}{\mathrm{M}} = \frac{3\mu \mathrm{Mg}}{\mathrm{M+m}}$

$\mathrm{a} = \frac{3\mu \mathrm{Mg}}{\mathrm{M+m}} > \frac{\mathrm{F}}{\mathrm{M}}$

Hence block and plate will move seperately and acceleration equal to its

maximum possible acceleration $\frac{\mu \mathrm{mg}}{\mathrm{M}}$

Q.3 Answer:

Circuit D with $\mathrm{G} = \frac{\mathrm{RS}}{\mathrm{R-S}}$

Solution:

Here, in figure D, current will flow through the circuit when key K_1 is closed and K_2 is open. The current flowing through the galvanometer is proportional to the deflection in it.

$I_1 = \frac{E}{R+G} = k\theta$

where, E-emf of the cell

R - resistance from the resistance box

G - galvanometer resistance for current I

θ - galvanometer deflection for current I

k - proportionality constant.

When K_2 is closed and by adjusting the shunt resistance S, we can make galvanometer deflection as $\theta/2$

Then the current in the circuit is:

$I_2 = \frac{E}{R+\frac{GS}{G+S}}$

Now, a fraction, $\left(\frac{S}{G+S}\right)$ of the current in the circuit is flows through the galvanometer, which is given by:

$I' = \frac{I_2 S}{G+S} = k\theta/2$

Now, from the above relations, we can get the resistance of the given galvanometer as:

$G = \frac{RS}{R-S}$

Q.4 Answer:

$\left[MLT^{-3}\right]$ and $\left[MLT^{-4}\right]$

Solution:

By principle of homogeneity, dimensiosns of each term must be same, i.e $[F] = [at] = \left[bt^2\right]$

So, $[a] = \left[\frac{F}{t}\right] = \frac{MLT^{-2}}{T} = MLT^{-3}$

$[b] = \left[\frac{F}{t^2}\right] = \frac{MLT^{-2}}{T^2} = MLT^{-4}$

Q.5 Answer:

101

Solution:

One of the earliest multiple-input gates to understand is the AND gate, because the output of gate will be high if and only all input are high. If any input is low the output is low.

From left, the first gate is OR gate and the output of OR gate is $Y_1 = A + B$ and above second gate is AND gate.

The output of AND gate is $Y = Y_1 . C = (A + B) . C$

For $101,\ Y = (1 + 0) . 1 = 1$

Q.6 Answer:

$g\left(1-\frac{1}{K}\right)$

Solution:

Suppose the stone is released from rest on the surface of the lake, then two forces will act on it. First, the force acting due to its weight downwards and second, the force acting upward due to the buoyancy force,

$K=\frac{\rho_{\mathrm{s}}}{\rho_{\mathrm{w}}}$

where ρ_{s} is density of stone and ρ_{w} is density of water.

If mass of stone is M, volume of stone is V and buoyancy force acting on it is F_{B}.

Stone is sinking, so net force will act downwards.

Net force on body, $F_{\mathrm{net}}=Mg-F_{\mathrm{B}}$

$F_{\mathrm{B}}=V\rho_{\mathrm{w}}g$ **and** $Mg=V\rho_{s}g$

$F_{\mathrm{net}}=V\rho_{\mathrm{s}}g\left(1-\frac{\rho_{\mathrm{w}}}{\rho_{\mathrm{s}}}\right)$

$ma=mg\left(1-\frac{\rho_{\mathrm{w}}}{\rho_{\mathrm{s}}}\right)=mg\left(1-\frac{1}{K}\right)$

Net acceleration, $a=g\left(1-\frac{1}{K}\right)$

Q.7 Answer:

$2.5\times10^{-3}\ \mathrm{rad\ s^{-1}}$

Solution:

$\mathrm{g'=g-\omega^2R\cos^2\lambda}\quad 0=\mathrm{g-\omega^2R\cos^2 60^\circ}$

$0=\mathrm{g}-\frac{\omega^2\mathrm{R}}{4}\ \Rightarrow\ \omega=2\sqrt{\frac{\mathrm{g}}{\mathrm{R}}}=\frac{1}{400}\ \mathrm{rad\ s^{-1}}=2.5\times10^{-3}\ \mathrm{rad\ s^{-1}}$

Q.8 Answer:

$\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

Solution:

According to Gauss law, $\phi_{\mathrm{total}}=\frac{q}{\varepsilon_0}$

Let electric flux linked with surfaces A, B and C are ϕ_{A}, ϕ_{B} and ϕ_{C} respectively. That is

$\phi_{\mathrm{total}}=\phi_{\mathrm{A}}+\phi_{\mathrm{B}}+\phi_{\mathrm{C}}$

Since $\phi_{\mathrm{C}}=\phi_{\mathrm{A}}$

$\therefore 2\phi_{\mathrm{A}}+\phi_{\mathrm{B}}=\phi_{\mathrm{total}}=\frac{q}{\varepsilon_0}$

or $\quad \phi_A=\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi_B\right)$

But $\phi_B=\phi\quad(\text{given})$

Hence, $\phi_A=\dfrac{1}{2}\left(\dfrac{q}{\varepsilon_0}-\phi\right)$

Q.9 Answer:

$T_h=T_c$

Solution:

Time period of any magnetic body place in an external magnetic field is:-

$$T = 2\pi\sqrt{\frac{I}{MB}}$$

$$T_{\text{hoop}} = 2\pi\sqrt{\frac{mR^2}{MB}} \ldots\ldots(1) I_{\text{hoop}} = mR^2$$

$$T_c = 2\pi\sqrt{\frac{\frac{mR^2}{2}}{\frac{M}{2}(B)}} \ldots\ldots(2) I_{\text{Solid cylinder}} = \frac{mR^2}{2}$$

From (1) and (2)

$T_h = T_c$

Q.10 Answer:

(i), (ii), (iii)

Solution:

The attraction force between nucleons is maximum, when they are separated at distance of $0.8\,\text{fm}$. When distance between two nucleons is less than $0.8\,\text{fm}$, then potential energy increases rapidly with distance, hence a strong repulsive force starts working. Nuclear force does not depend upon charge. When atomic mass lies between $30 - 170$, then due to saturation of nuclear forces, nuclear forces are short ranges.

Q.11 Answer:

MB

Solution:

Work done in rotating a bar magnet of magnetic moment M in a uniform magnetic field B is given by

$W = \Delta U = U_2 - U_1$

$\Rightarrow W = -MB\cos\theta_2 - (-MB\cos\theta_1)$

$[\because U = -MB\cos\theta]$

$\Rightarrow W = MB(\cos\theta_1 - \cos\theta_2)$

Now given,

$\theta_1 = 0^\circ$ (as magenet is in initially parallel to the magnetic field)

& $\theta_2 = 90^\circ$

$\therefore$ Putting in (i), We get

$W = MB(\cos 0^\circ - \cos 90^\circ)$

$\Rightarrow W = MB$

Q.12 Answer:

9 J

Solution:

Given: Force, $\overrightarrow{F} = \left(3\hat{\text{i}} + \hat{\text{j}}\right)\,\text{N}$

Initial position vector of particle, $\overrightarrow{r_1} = \left(4\hat{\text{i}} + 3\hat{\text{j}} - \widehat{\text{k}}\right)\,\text{m}$

Final position vector of particle, $\overrightarrow{r_1} = \left(2\hat{\text{i}} + \widehat{\text{k}}\right)\,\text{m}$

Displacement vector of particle is given as $\overrightarrow{s} = \overrightarrow{r_2} - \overrightarrow{r_1} = \left(2\hat{\text{i}} + 3\hat{\text{j}} - 2\widehat{\text{k}}\right)\,\text{m}$.

Work done by the force is given by

$W = \overrightarrow{F} \cdot \overrightarrow{s}$

$W = \left(3\hat{\mathrm{i}} + \hat{\mathrm{j}}\right) \cdot \left(2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - 2\widehat{\mathrm{k}}\right) = 6 + 3 = 9 \text{ J}$

Q.13 Answer:

2 and 3

Solution:

Every substance has a different value for the amount of heat absorbed or rejected to change the temperature of a unit mass by one unit. This quantity is referred to as the specific heat capacity of a body.

The specific heat is recognised as an intensive property that describes how much heat must be added to a particular substance to raise its temperature. It depends on the nature and temperature of the body.

Hence, statement 2 and 3 is correct.

Q.14 Answer:

-5 J

Solution:

From first law of thermodynamics,

$\Delta Q = \Delta W + \Delta U$

$\Delta U = 0$ **(process $ABCA$ is cyclic)**

$\therefore \Delta Q = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA}$

$5 = 10 + 0 + \Delta W_{CA}$

$\Rightarrow \Delta W_{CA} = -5J$

Q.15 Answer:

$\sqrt{\frac{2}{3}}$

Solution:

$c_{m:}c_{r.m.s} = \sqrt{\frac{2RT}{M}} : \sqrt{\frac{3RT}{M}}$

Q.16 Answer:

(A) is true but (R) is false

Solution:

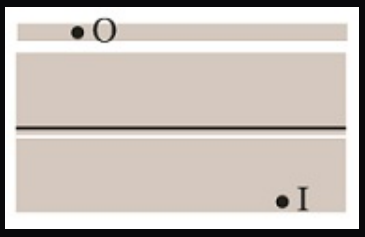
If a particle moves with uniform speed in a circle then its motion is said to be uniform circular motion. So assertion (A) is correct.

In uniform circular motion, magnitude of acceleration (also know as centripetal acceleration) remains constant $\left(a_c = \frac{v^2}{R}\right)$. But direction is always towards the centre. So acceleration vector is not constant. Hence, reason (R) is false.

Q.17 Answer:

If the lens used to obtain the image, the lens must be a converging lens and position of its principal focus can be found out

Solution:



If lens is used, then image is inverted and diminished i.e. magnification is less than 1. When converging lens is used, image is real. Magnification and difference between object and image is known so focus can be calculated for concave mirror (other kind of mirror is not possible).

Q.18 Answer:

10^{-34}

Solution:

It is given that a body of mass $m = 100\ \mathrm{g}$ moves at the speed of $v = 36\ \mathrm{km\ h^{-1}}$.

The de Broglie wavelength related to it,

$\lambda == \frac{h}{p} = \frac{h}{mv}$, here h is Planck's constant and p is momentum.

Substituting given values,

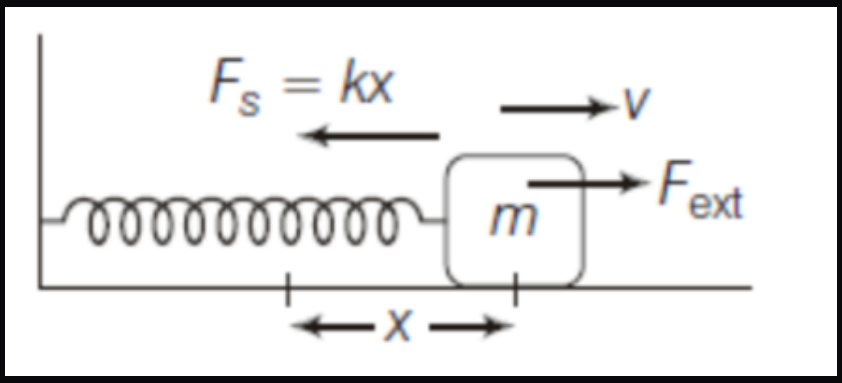
$\lambda = \frac{6.626\times10^{-34}}{0.1\times36\times\frac{1000}{3600}} \simeq 10^{-34}$

Q.19 Answer:

$W_{F_s} + W_{F_{ext}} > 0$

Solution:

Spring-force, $F_s = kx$



At $x = \frac{mg}{k}$

$F_s = kx = k\frac{mg}{k} = mg = F_{\text{ext}}$

So, $F_s = F_{ext}$

But at this point, the block must have gain some KE as its motion was accelerated.

So, $W_{F_s} + W_{F_{ext}} = \Delta\mathrm{KE}$

[from work-energy theorem]

$W_{F_s} + W_{F_{ext}} > 0$

Q.20 Answer:

$(-\hat{i} + \hat{j})$

Solution:

The magnetic field component is perpendicular to the direction of propagation and the direction of electric field.
Using vector algebra, $\vec{E} \times \vec{B}$ should be in z-direction
$\therefore \quad (\hat{i}+\hat{j}) \times (-\hat{i}+\hat{j}) = 2\hat{k}$

Q.21 Answer:

—

Solution:

$\mathrm{V}_{\mathrm{rms}}^2 = \frac{1}{\mathrm{T}} \int_0^{\mathrm{T}} \mathrm{V}^2 \mathrm{dt} = \frac{2}{\mathrm{T}} \int_0^{\frac{\pi}{2}} \mathrm{V}^2 \mathrm{dt}$
$\mathrm{T} = 2\mathrm{sec}$
$\mathrm{V} = 2\mathrm{t} \quad 0 \leq \mathrm{t} \leq 1$
$\mathrm{V}_{\mathrm{rms}}^2 = \frac{2}{\mathrm{T}} \int_0^1 4\mathrm{t}^2 \mathrm{dt}$
$= \int_0^1 4\mathrm{t}^2 \mathrm{dt}$
$= 4\frac{\mathrm{t}^3}{3}\Big]_0^1 = \frac{4}{3}$
$V_{\mathrm{rms}} = \sqrt{\frac{4}{3}}\ \mathrm{V}$

Q.22 Answer:

—

Solution:

As we know, $RD = \frac{w_1}{(w_1 - w_2)}$
Therefore,
$\frac{\Delta(RD)}{RD} \times 100 = \frac{\Delta w_1}{w_1} \times 100 + \frac{\Delta(w_1 - w_2)}{(w_1 - w_2)} \times 100$
$= \left(\frac{0.05}{5.00}\right) \times 100 + \left(\frac{0.05+0.05}{1.00}\right) \times 100$
$= 1 + 10 = 11\ \%$

Q.23 Answer:

—

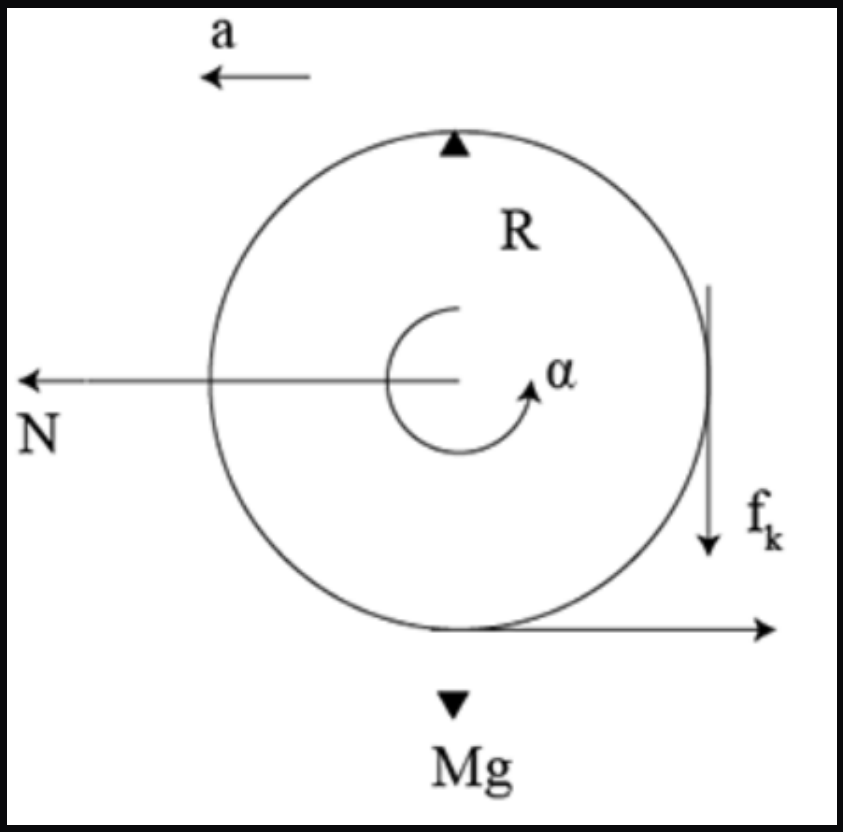
Solution:

For translation, $N - f_{\mathrm{s}} = ma$ for rotation $(f_s - f_k)\, R = I\alpha$ and $a = R\alpha$.
Solving, $f_k = 0.8\ \mathrm{N}; f_{\mathrm{s}} = 1.4\ \mathrm{N}$

$\Rightarrow \mu = \dfrac{P}{10} = \dfrac{f_k}{2\ \mathrm{N}}$

$\Rightarrow \dfrac{P}{10} = \dfrac{0.8}{2}$

$\Rightarrow P = 4$



Q.24 Answer:

—

Solution:

Let the length of various edges in increase order be $l, x, 2l$ respectively

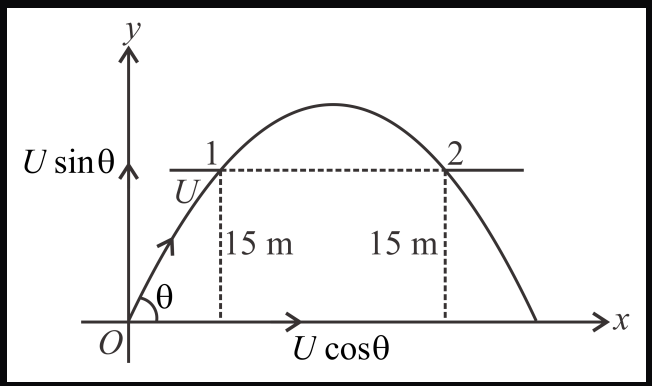
$R_{\max} = \frac{\rho 2l}{xl} = \frac{2\rho}{x}; R_{\min} = \frac{\rho l}{2lx} = \frac{\rho}{2x}$

$\therefore \ \frac{R_{\max}}{R_{\min}} = 4$

Q.25 Answer:

—

Solution:



The ball is projected with the velocity $u = 52\ \mathrm{m\ s^{-1}}$ at angle θ with horizontal.

We have to calculate the time for which the ball is at least $15\ \mathrm{m}$ above the ground.

Using the equation of motion in the vertical direction

$y = u(\sin\theta)t - \frac{1}{2}gt^2$.

Vertical displacement $y = 15$ m, the vertical velocity is $u\sin\theta$.

$15 = 52 \times \frac{5}{13}t - \frac{1}{2} \times 10t^2 \left[\text{If } \tan\theta = \frac{5}{12}, \text{ then } \sin\theta = \frac{5}{13}\right]$

$5t^2 - 20t + 15 = 0$

$t^2 - 4t + 3 = 0$

$t^2 - 3t \times t + 3 = 0$

$t(t-3) - 1(t-3) = 0$

$t_1 = 1\text{ s} \Rightarrow t_2 = 3\text{ s}.$

At $t = 1$ s and $t = 3$ s ball is at a height of minimum 15 m from ground.

So, the time for which ball is at least 15 m above the ground is $t_2 - t_1 = 3 - 1 = 2$ s.

Q.26 Answer:

$73.08\ \text{kPa}$

Solution:

$\mathrm{P} = \mathrm{X_A P_A^0} + \mathrm{x_B P_B^0}$

$\mathrm{P^0_{Heptane}} = 105.2\ \text{kPa}$

$\mathrm{P^0_{Octane}} = 46.8\ \text{kPa}$

$\mathrm{P_{Total}} = \mathrm{P^0_{Heptane} X_{Heptane}} + \mathrm{P^0_{Octane} X_{Octane}}$

$= 105.2 \times \left(\frac{\frac{26}{100}}{\frac{26}{100}+\frac{35}{114}}\right) + 46.8 \times \left(\frac{\frac{35}{114}}{\frac{26}{100}+\frac{35}{114}}\right)$

$= 73.08\ \text{kPa}$

Q.27 Answer:

Magnetic behaviour

Solution:

Geometry of complexes $\rightarrow$ Both square planar

Hybridisation of central metal cation $\rightarrow \mathrm{dsp^2}$

Magnetic behaviour $\left[\mathrm{Cu(en)_2}\right]^{2+} \rightarrow$ Paramagnetic; $\left[\mathrm{Ni(dmg)_2}\right] \rightarrow$ Diamagentic

Number of stereoisomers $= 0$

Q.28 Answer:

$\left(\mathrm{CH_3}\right)_3\mathrm{CBr}$

Solution:

Tert-butyl carbocation is more stable than benzyl carbocation. 9 alpha hydrogen stability dominates over resonance due to benzene ring.

Q.29 Answer:

F > N > C > B > Si

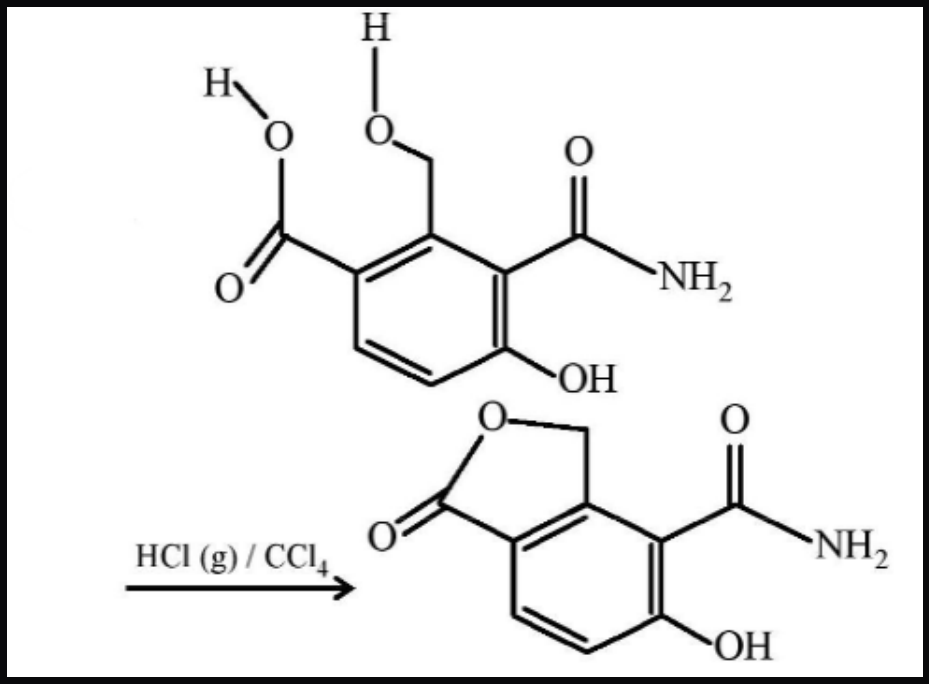
Solution:

Non-metallic character in a group decreases from top to bottom but it increases in a period when we move from left to right. This is because ionisation energy increases in a period and decreases in a group. Therefore, among B, C, N, F and Si, the non-metallic character decreases in the order F > N > C > B > Si.

Q.30 Answer:



Solution:



Q.31 Answer:

1/10

Solution:

Indicator is weak organic acid HA

$\mathbf{pH = pK_a + \log \frac{[A^-]}{[HA]}}$

$\mathbf{6.5 = 7.5 + \log \frac{[A^-]}{[HA]}}$

$\mathbf{\log \frac{[A^-]}{[HA]} = -1}$

$\mathbf{\therefore \frac{[A^-]}{[HA]} = 0.10}$

Q.32 Answer:

$T - \frac{2}{3 \times 0.082}$

Solution:

T' is final temperature.

$q = \Delta E - W$

$= C_V(T' - T) + P_{ext}(2 - 1)$

$= C_V(T' - T) + 1\ L\ atm$

$q = \frac{3}{2}R\ (T' - T) + 1\ L\ atm = 0$

$T' - T = -\frac{2}{3}\frac{1}{R}$

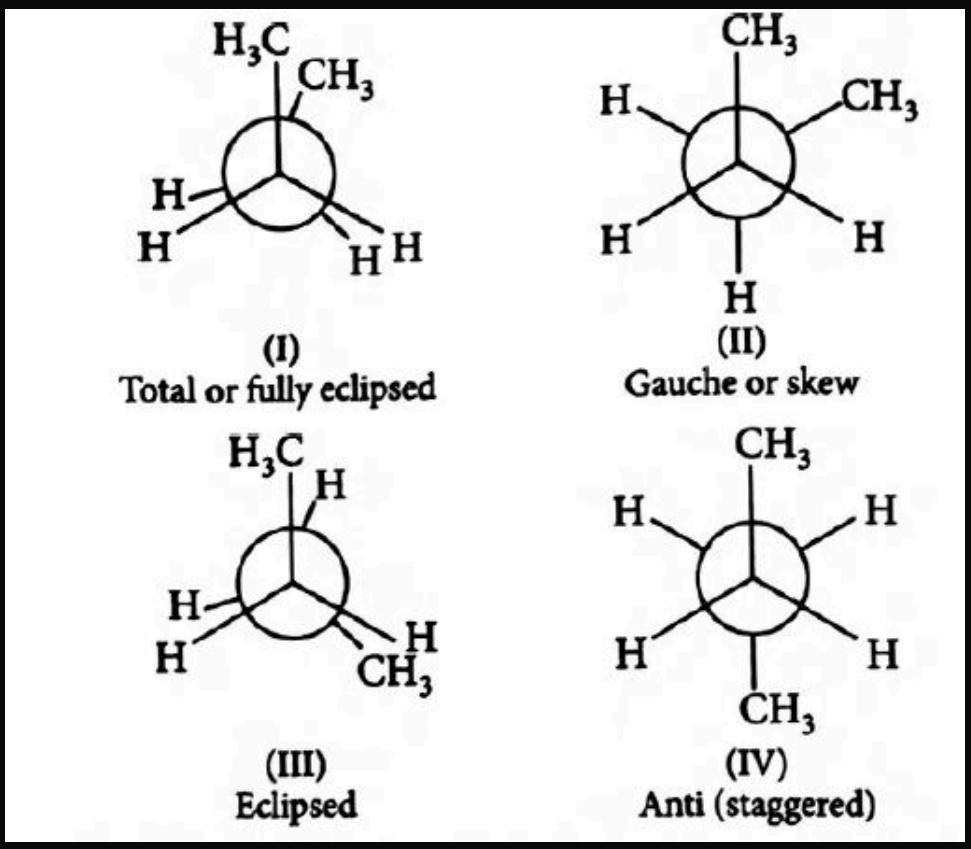
$T' = T - \frac{2}{3R}$

$= T - \frac{2}{3\times 0.082}$

Q.33 Answer:

Staggered < Gauche < Eclipsed < Fully eclipsed

Solution:



Fully eclipsed conformation is least stable because of maximum torsional and Van der waal's repulsion so, it has maximum energy. Eclipsed conformation has lesser repulsion in comparison to fully eclipsed conformation therefore, has more stability and hence lesser energy. Gauche conformation has methyl groups at 60° from each other whereas staggered conformation has methyl groups at 180° from each other therefore, staggered is more stable than gauche due to less repulsion and hence lesser energy than gauche. Thus overall, order of energy is : Staggered < Gauche < Eclipsed < Fully eclipsed

Q.34 Answer:

Potassium dichromate

Solution:

Potassium dichromate $K_2Cr_2O_7$ is an orange-red crystalline soluble solid substance. It is a good oxidizing agent and is used in making chrome pigments, azo dyes and as a bleaching agent.

Q.35 Answer:

$130\ S\ cm^2\ mol^{-1}$

Solution:

The Kohlrausch's law states that, the molar conductivity of an electrolyte at infinite dilution, is equal to the sum of contributions of the molar conductivities of individual ions and it is independent of migration of ions.

$\Lambda_m^\infty = (57 + 73)\ \text{S cm}^2\ \text{mol}^{-1} = 130\ \text{S cm}^2\ \text{mol}^{-1}$

Q.36 Answer:

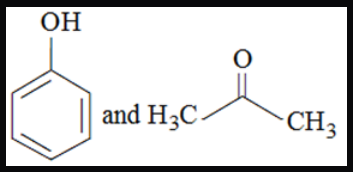
The H - O - H bond angle in H_2O is larger than the H - C - H bond angle in CH_4 .

Solution:

Bond Angle	Molecule
104.5^o	H_2O
107°	NH_3
$109^o28'$	CH_4

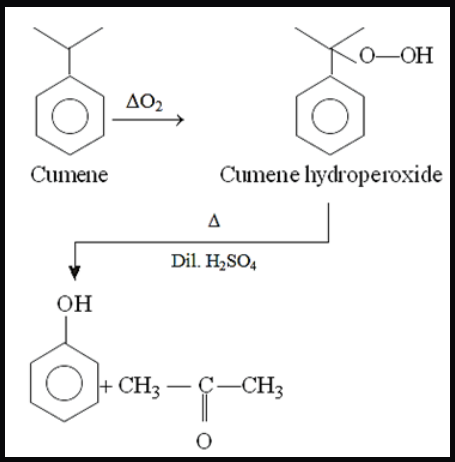
All the molecules are sp^3 hybridized and Bond angle of H_2O is smaller than NH_3 because in H_2O two lone pairs are present due to which there will be more lone pair-lone pair repulsion while inNH_3,only one lone pair is present, so lone pair-bond pair repulsion is present only.

Q.37 Answer:



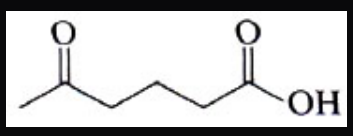
Solution:

Cumene hydroperoxide reaction

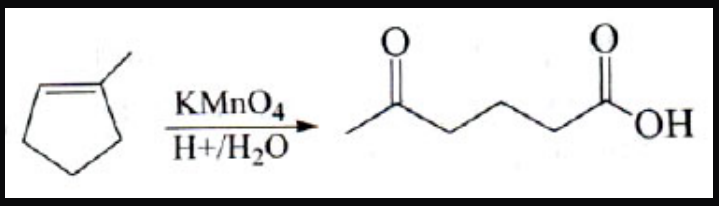


This is commercial method of preparation of phenol.

Q.38 Answer:



Solution:



Q.39 Answer:

$6 \cdot 022 \times 10^{20}$

Solution:

For HCl**,** $\mathrm{n}-$**factor is 1**

So, for HCl**,**

Normality $=$ **Molarity**

$\text{Molarity} = \frac{\text{Number of moles of HCl}}{\text{Total voume (in L)}}$ $\text{Number of moles of HCl} = \text{Molarity} \times \text{Total volume (in L)}$

In 1 mol **of** HCl**, the number of** HCl **molecules present** $= \mathrm{N_A}$ $=$ **Avogadro's number**

So, the number of HCl **molecules in** 10^{-3} **moles**

$= 10^{-3} \times \mathrm{N_A} \ = 10^{-3} \times 6.022 \times 10^{23} \ = \ 6.022 \times 10^{20}$ molecules

Q.40 Answer:

The volume on the reaction vessel was decreased

Solution:

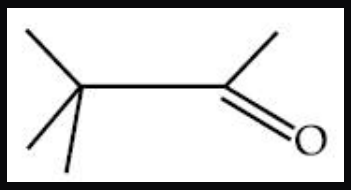
Volume of the vessel was decreased, which led to sudden increase in $[\mathrm{H_2}]$**. The reaction then moves in direction to achieve a state of new equilibrium.**

A catalyst does not affect the equilibrium concentration of reactants and products.

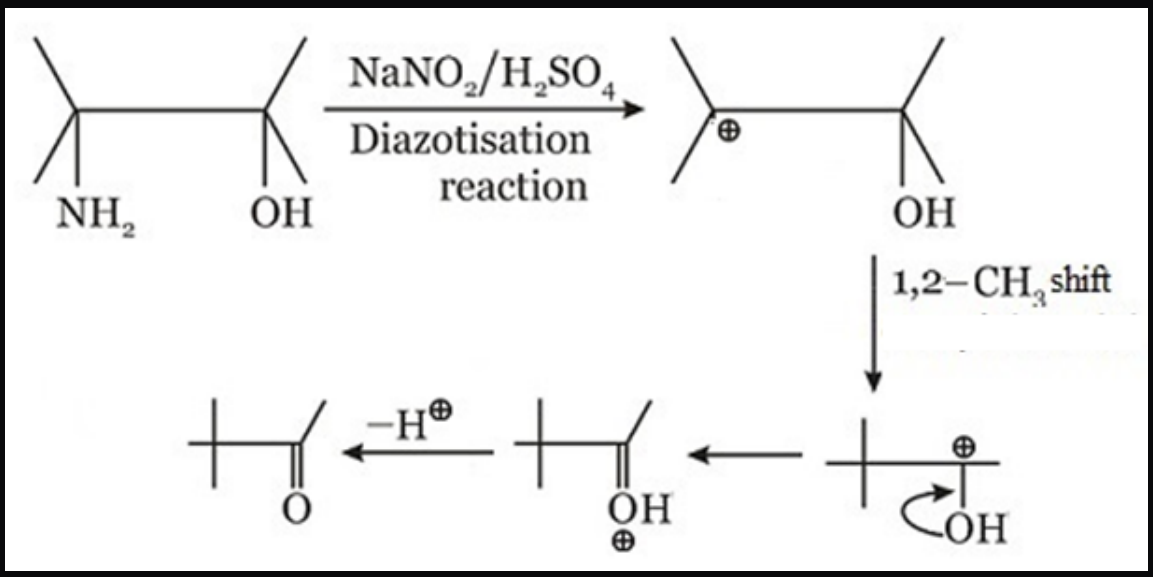
Increase in pressure would decrease concentration as reaction favoured forward.

$\mathrm{P_4(\ s)}$ **will have no effect on** $\mathrm{k_p}$ **and concentration of** $\mathrm{H_2(\ g)}$ **(because active mass of solid is taken to be unit)**

Q.41 Answer:



Solution:



Q.42 Answer:

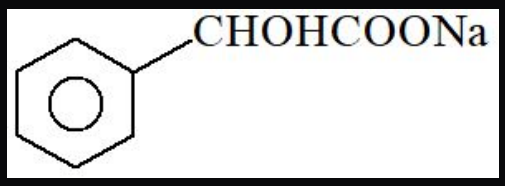
Assertion is true but Reason is false.

Solution:

Bond	Bond energy
$Si-Si$	$340\ kJ/mole$
$Si-O$	$452\ kJ/mole$

Thus, $Si-O$ bond is stronger, owing to the higher overlap due to the small size of the oxygen atom.
Silicon cannot form $Si=Si$ due to poor overlapping of $p\pi-d\pi$ bond. Si has no tendency to form double bonds with itself due to its large size. The bond formation is difficult due to the large size.

Q.43 Answer:



Solution:

It is an example of internal Cannizzaro reaction; where -CHO is oxidized while ketonic group of the same molecule is reduced. Since keto group can't be oxidized, alcohol corresponding to -CHO group will not be formed.

Q.44 Answer:

(p)-3, (q)-4, (r)-1, (s)-1

Solution:

A. **Nucleic acids are long polymers in which monomeric units are nucleotides.**
B. **RNA has the base uracil in place of thymine.**
C. **DNA has the base thymine paired with adenine.**
D. **DNA has double-helix structure. The two strands of DNA helix are said to be complementary to each other in the sense that the sequence of bases in one strand automatically determines that of order. Base pairs in DNA are held by H-bonds denoted by dotted lines.**

Q.45 Answer:

Both (a) and (c).

Solution:

The outer electron experiences two opposing forces:
Force of attraction from nucleus.
Force of repulsion from inner electrons.
Screening effect or shielding effect can be defined as a reduction in the nuclear charge on the electron cloud.
Screening effect or shielding effect is the decrease in the force of attraction exerted by the nucleus on the valence electrons due to electrons in the inner shells.
The higher the screening effect on the outermost electrons causes less attraction from the nucleus and can be easily removed, leading to the lower value of ionization potential.
An atom's electron affinity is a measure of the attraction that exists between the nucleus (positively charged) and the electron (negatively charged) implying the factors that tend to reduce this attraction will also reduce electron affinity.

The outer electrons and the nucleus's sheilding effect increases on the increase in number of electrons in the inner shells. Therefore, the electron affinity value of the elements decreases with the increasing shielding or screening effect.

Q.46 Answer:

—

Solution:

Weight of organic compound = 29.5 mg

$NH_3 + HCl \rightarrow NH_4Cl$

$HCl + NaOH \rightarrow NaCl + H_2O$

(Remaining) $15 \times 0.1\ M$

= 1.5 mmol

Total millimole of HCl = 2

Millimole used by $NH_3 = 2 - 1.5 = 0.5$

Weight of NH_3 **= 0.5 × 17 mg = 8.5 mg**

Weight of nitrogen $= \frac{14}{17} \times 8.5mg = 7mg$

$\%\ \text{Nitrogen} = \frac{7}{29.5} \times 100 = 23.7\%$

Q.47 Answer:

—

Solution:

$\Delta T_f = iK_f m$ **(i =** 1 **for non-electrolyte)**

$2 = 0.76 \times \frac{n}{\frac{100}{1000}}$

$n = \frac{2}{7.6}$

Relative lowering in vapour pressure = X_B **(mole fraction of solute)**

$\frac{\Delta P}{P_A^0} = \frac{2/7.6}{100/18}$

$\frac{\Delta P}{760} = \frac{36}{760}$

$\Delta P = 36$

$P = 760 - 36$

$P = 724\ \text{mm of Hg}$

Q.48 Answer:

—

Solution:

Let x be the pressure of C_2H_4msub **decompose after 20 minutes of decompostion.**

Then, $80 - x + 2x = 120$**,**

x = 40 torr = 50 % of initial pressure

Here $t_{1/2} = 20$ **min.**

For 75 % reaction, fraction left $= \frac{25}{100} = \frac{1}{4} = \left(\frac{1}{2}\right)^2$

No. of half lives = 2.

Time needed for 75% reaction $= 2 \times 20 = 40$ **min.**

Q.49 Answer:

—

Solution:

Transition energy $= [-2.41 - (-5.42)] \times 10^{-12}\text{erg}$

$= 3.01 \times 10^{-12}\text{erg}$

$= 3.01 \times 10^{-19}\text{J} \quad [\because 1\text{erg} = 10^{-7}\text{J}]$

Also, $\Delta\text{E} = \frac{\text{hc}}{\lambda}$

$\Rightarrow \quad \lambda = \frac{6.625\times10^{-34}\times3\times10^{8}}{3.01\times10^{-19}}\text{m}$

$= 660 \times 10^{-9}\text{m}$

= 660 nm

Q.50 Answer:

—

Solution:

$3\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 5\text{Fe}^{2+} + 24\text{H}^+ \longrightarrow 3\text{Mn}^{2+} + 10\text{CO}_2 + 5\text{Fe}^{3+} + 12\text{H}_2\text{O}$

$\because$ Number of moles of MnO_4^- required to oxidise 5 moles of $C_2O_4^{2-} = 3$

$\therefore$ Number of moles of MnO_4^- required to oxidise 1 mole of $\text{C}_2\text{O}_4^{2-} = \frac{3}{5} = 0.6$ mol

hence, the number of moles Of MnO_4 required to oxidise 5 moles of ferrous oxalate. $= 0.6 \times 5 = 3$ mol

Q.51 Answer:

$\frac{9!}{2!\,4!}$

Solution:

C	I	N	O	R	S	U
1	2	2	1	1	1	1

Total number of arrangements $= \frac{9!}{2!2!}$

The number of ways, vowels in these words are arranged

$= \frac{4!}{2!}$

$\begin{pmatrix} I & O & U \\ 2 & 1 & 1 \end{pmatrix}$

But the vowels are restricted to have only one arrangement i.e. IIOU

$\Rightarrow$ Total number of possible arrangement of the letters of the word INCURSION $= \frac{9!/2!2!}{4!/2!} = \frac{9!}{2!4!}$

Q.52 Answer:

$\text{a} = -3$ and $\text{b} = 9/2$

Solution:

$\lim\limits_{x\to0}\left(\frac{\sin\,3x}{x^3} + \frac{a}{x^2} + b\right) = \lim\limits_{x\to0}\frac{\sin\,3x+ax+bx^3}{x^3}$

$= \lim\limits_{x\to0}\frac{3\frac{\sin\,3x}{3x}+a+bx^2}{x^2}$

limit exists if

$3 + a = 0$

or $a = -3$

$\therefore \quad L = \lim\limits_{x\to0}\frac{\sin\,3x-3x+bx^3}{x^3} = 27\left(\lim\limits_{t\to0}\frac{\sin\,t-t}{t^3} + b\right) = 0 \quad \left(\textbf{Putting } 3x = t\right) = -\frac{27}{6} + b = 0$

or $b = \frac{9}{2}$

Q.53 Answer:

0

Solution:

$I = 2\int_0^1 [x\sin(\pi x)]dx$

Now, $x\sin(\pi x) \in (0,1)$

as $x \in (0,1)$

$\therefore\ [x\sin(\pi x)] = 0$

$\therefore\ I = 2\int_0^1 0dx = 0$

Q.54 Answer:

$\frac{9}{16}$

Solution:

$P(H) = \frac{3}{4}, P(T) = \frac{1}{4}$

$P\left(\frac{x\geq 6}{x>3}\right) = \frac{P(x\geq 6 \cap x>3)}{P(x>3)} = \frac{P(x\geq 6)}{P(x\geq 4)}$

$P(x \geq 6) = \left(\frac{3}{4}\right)^5\frac{1}{4} + \left(\frac{3}{4}\right)^6\frac{1}{4} + \left(\frac{3}{4}\right)^7\left(\frac{1}{4}\right) + \ldots = \frac{\left(\frac{3}{4}\right)^5\frac{1}{4}}{1-\frac{3}{4}} = \left(\frac{3}{4}\right)^5$

$P(x \geq 4) = \left(\frac{3}{4}\right)^3\left(\frac{1}{4}\right) + \left(\frac{3}{4}\right)^4\left(\frac{1}{4}\right) + \left(\frac{3}{4}\right)^5\left(\frac{1}{4}\right) + \ldots\ldots = \frac{\left(\frac{3}{4}\right)^3\left(\frac{1}{4}\right)}{1-\frac{3}{4}} = \left(\frac{3}{4}\right)^3$

$P\left(\frac{x\geq 6}{x>3}\right) = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$

Q.55 Answer:

centre of ellipse

Solution:

Let, $P' \equiv (a\cos\theta, b\sin\theta)$

then,

$Q' \equiv \left(a\cos\left(\theta + \frac{2\pi}{3}\right),\ b\sin\left(\theta + \frac{2\pi}{3}\right)\right)$

$R' \equiv \left(a\cos\left(\theta + \frac{4\pi}{3}\right),\ b\sin\left(\theta + \frac{4\pi}{3}\right)\right)$

Let centroid of $\Delta P'Q'R' \equiv (x', y')$

$x' = a\left[\frac{\cos\theta + \cos\left(\theta+\frac{2\pi}{3}\right) + \cos\left(\theta+\frac{4\pi}{3}\right)}{3}\right]$

$= \frac{a}{3}\left[\cos\theta + 2\cos\left(\theta+\pi\right)\cos\frac{\pi}{3}\right]$

$= \frac{a}{3}\left[\cos\theta + 2(-\cos\theta)\left(\frac{1}{2}\right)\right]$

$= 0$

And, $y' = b\left[\frac{\sin\theta + \sin\left(\theta+\frac{2\pi}{3}\right) + \sin\left(\theta+\frac{4\pi}{3}\right)}{3}\right]$

$= b\left[\frac{\sin\theta + 2\sin(\theta+\pi)\cos\left(\frac{\pi}{3}\right)}{3}\right]$

$= b\left[\frac{\sin\theta + 2(-\sin\theta)\left(\frac{1}{2}\right)}{3}\right]$

$= 0$

Hence, centroid of triangle coincides with the centre of ellipse.

Q.56 Answer:

reflexive and symmetric

Solution:

Here, the relation on N is

$S = \{(x, y) \mid y/2 \leq x \leq 2y\}$

Let $x \in S$, then $\frac{x}{2} \leq x \leq 2x \Rightarrow xSx$

$\Rightarrow S$ is reflexive.

Now, let $x, y \in S$ such that xSy

$\Rightarrow \frac{y}{2} \leq x \leq 2y$

$\because \frac{y}{2} \leq x$ and $x \leq 2y$

$\Rightarrow y \leq 2x$ and $\frac{x}{2} \leq y$

$\Rightarrow \frac{x}{2} \leq y \leq 2x \Rightarrow ySx$

$\Rightarrow S$ is symmetric.

Let $x, y, z \in S$ such that

xSy and ySz

$\frac{y}{2} \leq x \leq 2y$ and $\frac{z}{2} \leq y \leq 2z$

Now, $x \leq 2y$

$\leq 2 \times 2z$

$\leq 4z \leq 2z$

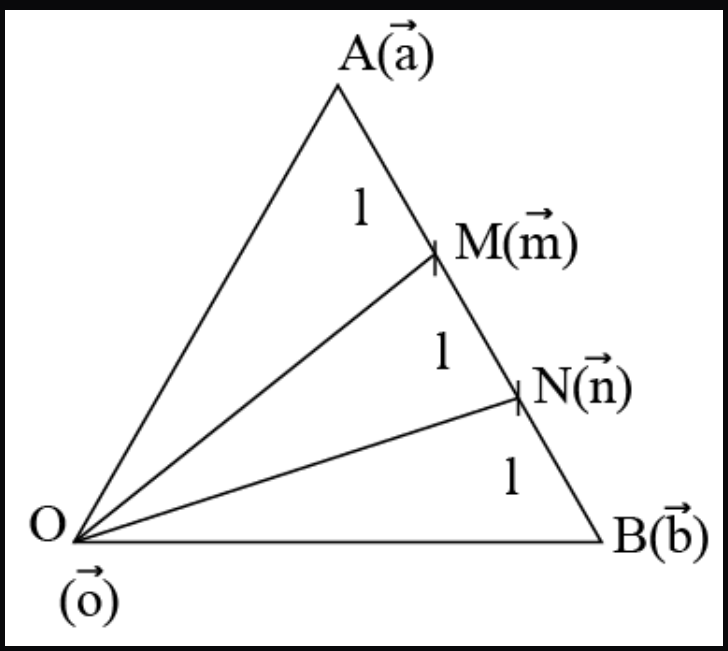
x is not related to z.

$\Rightarrow S$ is not transitive.

Q.57 Answer:

$\frac{13}{18}$

Solution:



$\vec{m} = \frac{2\vec{a}+\vec{b}}{3}, \vec{n} = \frac{\vec{a}+2\vec{b}}{3}$

$\vec{m} \cdot \vec{n} = \frac{1}{9}(2\vec{a} + \vec{b}) \cdot (\vec{a} + 2\vec{b})$

$= \frac{1}{9}\left[2|\vec{a}|^2 + 2|\vec{b}|^2 + 5\vec{a} \cdot \vec{b}\right]$

$= \frac{1}{9}\left[2 + 2 + 5 \times 1 \times 1 \times \frac{1}{2}\right] = \frac{13}{18}$

Q.58 Answer:

1

Solution:

Put $x = y = 1$

$f^2(1) + f^2(3) = 2f(1)$

$\Rightarrow (f(1) - 1)^2 = 0 \Rightarrow f(1) = 1$

$f(x)f(1) + f(\frac{3}{x})f(3) = 2f(x)$

$\Rightarrow f(x) = f(\frac{3}{x}) \ \forall x > 0 \quad \ldots(1)$

$\Rightarrow f(x)f(\frac{3}{x}) + f(\frac{3}{x})f(x) = 2f(3)$

$\Rightarrow f(x)f(\frac{3}{x}) = 1 \quad \ldots.(2)$

$\Rightarrow$ **from** (1) **and** (2)

$f^2(x) = 1 \ \forall x > 0$

Put $x = y = \sqrt{t}$

$f^2\left(\sqrt{t}\right) + f^2\left(\frac{3}{\sqrt{t}}\right) = 2f\left(t\right)$ **or** $f(t) > 0$

$\Rightarrow f(x) = 1 \ \forall x > 0$

Q.59 Answer:

4

Solution:

Let, the position vectors are $A\left(\vec{a}\right), B\left(\vec{b}\right), C\left(\vec{c}\right), D\left(\vec{d}\right)$

So, $\Delta ABC = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right|$

$= \frac{1}{2}\left|\left(\vec{b} - \vec{a}\right) \times \left(\vec{c} - \vec{a}\right)\right|$ **...** (1)

Now, $\overrightarrow{AB} \times \overrightarrow{CD} = \left(\vec{b} - \vec{a}\right) \times \left(\vec{d} - \vec{c}\right)$

$= \vec{b} \times \vec{d} - \vec{b} \times \vec{c} - \vec{a} \times \vec{d} + \vec{a} \times \vec{c}$

$\overrightarrow{BC} \times \overrightarrow{AD} = \left(\vec{c} - \vec{b}\right) \times \left(\vec{d} - \vec{a}\right)$

$= \vec{c} \times \vec{d} - \vec{c} \times \vec{a} - \vec{b} \times \vec{d} + \vec{b} \times \vec{a}$

$\overrightarrow{CA} \times \overrightarrow{BD} = \left(\vec{a} - \vec{c}\right) \times \left(\vec{d} - \vec{b}\right)$

$= \vec{a} \times \vec{d} - \vec{a} \times \vec{b} - \vec{c} \times \vec{d} + \vec{c} \times \vec{b}$

$\left|\overrightarrow{AB} \times \overrightarrow{CD} + \overrightarrow{BC} \times \overrightarrow{AD} + \overrightarrow{CA} \times \overrightarrow{BD}\right| = 2\left|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right|$

$= 2\left|\left(\vec{b} - \vec{a}\right) \times \left(\vec{c} - \vec{a}\right)\right|$

$= 4\Delta ABC$

$\Rightarrow \lambda = 4$

Q.60 Answer:

150

Solution:

Total number of functions $= 3^5$

(since each of $1,2,3,4$ **or** 5 **can correspond to any of** a, b **or** c**)**

The number of functions that corresponds to only one element of B **is** ${}^3C_1 \times 1^5$ **and the number of functions that**

correspond to atmost two elements of B is $^3C_2 \times 2^5$.

Total number of onto functions $= 3^5 - {}^3C_2 \times 2^5 + {}^3C_1 \times 1^5 = 243 - 96 + 3 = 150$

$\left({}^3C_1 \times 1^5 \text{ is repeated twice in } {}^3C_2 \times 2^5\right)$

Q.61 Answer:

Increasing for all $x \in (0, 1)$

Solution:

$f(x) = \int_0^x \left(5\,In\left(1+t^2\right) - 10t\,\tan^{-1}t + 16\sin t\right)dt$

$\Rightarrow f\prime(x) = 5\,In\left(1+x^2\right) - 10x\,\tan^{-1}x + 16\sin\,x$

$\Rightarrow f\prime\prime(x) = 2\left(8\cos x - 5\tan^{-1}x\right)$

$\Rightarrow f\prime\prime\prime(x) = -2\left(8\sin x + \frac{5}{1+x^2}\right) < 0 \forall x \in (0,1)$

So, $f\prime\prime(x)$ is decreasing $\forall x \in (0,1)$

$\Rightarrow f\prime\prime(x) > f\prime\prime(1) = 2\left(8\cos 1 - \frac{5\pi}{4}\right) > 2\left(8\cos\frac{\pi}{3} - \frac{5\pi}{4}\right)$

$= 2\left(4 - \frac{5\pi}{4}\right) > 0$

So, $f\prime(x)$ is increasing, for $x > 0$. $f\prime(x) > f\prime(0) = 0$

So, $f(x)$ is increasing , for $x > 0, f(x) > f(0) = 0$

So, $\int_0^x f(t)$ **is positive and increasing.**

Q.62 Answer:

$b, \sqrt{a}, c$ are in geometric progression

Solution:

$a = 1 + 10 + 10^2 + \ldots\ldots\ldots + 10^{54} = \frac{10^{55}-1}{9}$

$b = \frac{10^5-1}{9}$

$c = \frac{\left(10^5\right)^{11}-1}{10^5-1} = \frac{10^{55}-1}{10^5-1}$

$\Rightarrow \frac{a}{b} = c$

$\Rightarrow a = bc$

$\Rightarrow b, \sqrt{a}, c$ **are in G.P.**

Q.63 Answer:

$\frac{343}{9^9}$

Solution:

Total cases $= 9^9$

All diagonal elements are zero and $a_{ij} = -a_{ji}$

So favourable cases $= 7^3$

Hence, required probability $= \frac{7^3}{9^9}$

Q.64 Answer:

If both the statements are true and statement-2 is the correct explanation of statement-1.

Solution:

Eliminating n we get,

$\lambda(l+m)^2+lm=0 \Rightarrow \frac{\lambda l^2}{m^2}+(2\lambda+1)\frac{l}{m}+\lambda=0 \Rightarrow \frac{l_1l_2}{m_1m_2}=1.$

Eliminating m,

$\Rightarrow \lambda l^2-ln-n^2=0 \Rightarrow \frac{l_1l_2}{n_1n_2}=\frac{-1}{\lambda}$

$l_1l_2+m_1m_2+n_1n_2=0 \Rightarrow 1+1-\lambda=0 \Rightarrow \lambda=2.$

Q.65 Answer:

none of these

Solution:

One of them has to be maxima and other minima

Q.66 Answer:

$70\left(\frac{1-t}{1+t}\right)^4$

Solution:

$$\left[\left(t^{-1}-1\right)x+\left(t^{-1}+1\right)^{-1}x^{-1}\right]^8$$

$$=\left[\left(\frac{1}{t}-1\right)x+\left(\frac{1}{t}+1\right)^{-1}\frac{1}{x}\right]^8$$

Let T_{r+1} be the term independent of x, then

$\mathrm{T}_{r+1}={}^8\mathrm{C}_r\left(\frac{1}{t}-1\right)^{8-r}x^{8-r}\cdot\left(\frac{1}{t}+1\right)^{-r}\left(\frac{1}{x}\right)^r$

$={}^8\mathrm{C}_r\left(\frac{1}{t}-1\right)^{8-r}\cdot\left(\frac{1}{t}+1\right)^{-r}\cdot x^{8-2r}$

$\therefore 8-2r=0 \Rightarrow r=4$

$\therefore \mathrm{T}_5$ **is the term independent of x and**

$$\mathrm{T}_5={}^8C_4\left(\frac{1}{t}-1\right)^4\cdot\left(\frac{1}{t}+1\right)^{-4}$$

$$={}^8\mathrm{C}_4\left(\frac{1-t}{t}\right)^4\cdot\left(\frac{1+t}{t}\right)^{-4}$$

$$={}^8\mathrm{C}_4\left(\frac{1-t}{1+t}\right)^4=\frac{8\cdot7\cdot6\cdot5}{4\cdot3\cdot2\cdot1}\left(\frac{1-t}{1+t}\right)^4$$

$$=70\cdot\left(\frac{1-t}{1+t}\right)^4$$

Q.67 Answer:

$\frac{1}{2}\left(1-\frac{1}{\sqrt{e}}\right)$ sq. units

Solution:

To maximize $y(x)=x\,e^{-x^2}$

$y'(x)=e^{-x^2}+x\;\;e^{-x^2}(-2x)$

$\qquad=e^{-x^2}\left[1-2x^2\right]$

$y'(x)=0 \Rightarrow \;\; x=\pm\frac{1}{\sqrt{2}}$ **are the critical points.**

$y''(x)=e^{-x^2}[-4x]+\left(1-2x^2\right)\left(e^{-x^2}\right)(-2x)$

$= e^{-x^2}\left[-4x - 2x + 4x^3\right]$

$= e^{-x^2}\left[4x^3 - 6x\right]$

$= 2xe^{-x^2}\left[2x^2 - 3\right]$

$y'\left(\frac{1}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}}\ e^{-\frac{1}{2}}\left[\frac{2}{2} - 3\right] < 0$

$\therefore\ \ x = \frac{1}{\sqrt{2}}$ **is a point of maximum.**

∴Required area $= \int_0^{\frac{1}{\sqrt{2}}} x\, e^{-x^2} dx$

Put $-x^2 = t$**,**

$-2xdx = dt \Rightarrow xdx = \frac{-1}{2}dt$

$= -\frac{1}{2}\int_0^{-\frac{1}{2}} e^t\, dt$

$= \frac{1}{2}\left|e^t\right|_0^{-1/2} = \frac{-1}{2}\left[\frac{1}{\sqrt{e}} - 1\right]$

$= \frac{1}{2}\left(1 - \frac{1}{\sqrt{e}}\right)$ sq. units.

Q.68 Answer:

$f(x) = \frac{2x}{2-x^2}$

Solution:

The given differential equation is,

$y(1 + xy)dx - xdy = 0$

$\Rightarrow (ydx - xdy) + xy^2dx = 0$

$\Rightarrow \frac{ydx - xdy}{y^2} + x\ dx = 0$

$\Rightarrow d\left(\frac{x}{y}\right) + x\ dx = 0$

On integrating both sides, we get

$\int d\left(\frac{x}{y}\right) + \int xdx = 0$

$\Rightarrow \frac{x}{y} + \frac{x^2}{2} = C$

If the above curve passes through the point $(1,\ 2)$**, then,**

$\frac{1}{2} + \frac{1}{2} = C \Rightarrow C = 1$

The given curve is $\frac{x}{y} + \frac{x^2}{2} = 1$

$\therefore\ y = \frac{2x}{2-x^2}$

$\Rightarrow f(x) = \frac{2x}{2-x^2}$

Q.69 Answer:

$\sqrt{3}$

Solution:

Given, $\frac{|x+(y+1)i|}{|x+(y-1)i|} = \sqrt{3}$

$\Rightarrow \quad x^2 + (y+1)^2 = 3\left[x^2 + (y-1)^2\right]$

$\Rightarrow \quad x^2 + y^2 - 4y + 1 = 0$

On comparing with

$x^2 + y^2 + 2gx + 2fy + c = 0$**, we get**

$g = 0, \quad f = -2, \quad c = 1$

∴ Radius of circle= $\sqrt{g^2 + f^2 - c} = \sqrt{(-2)^2 - 1} = \sqrt{3}$

Q.70 Answer:

1 : 4

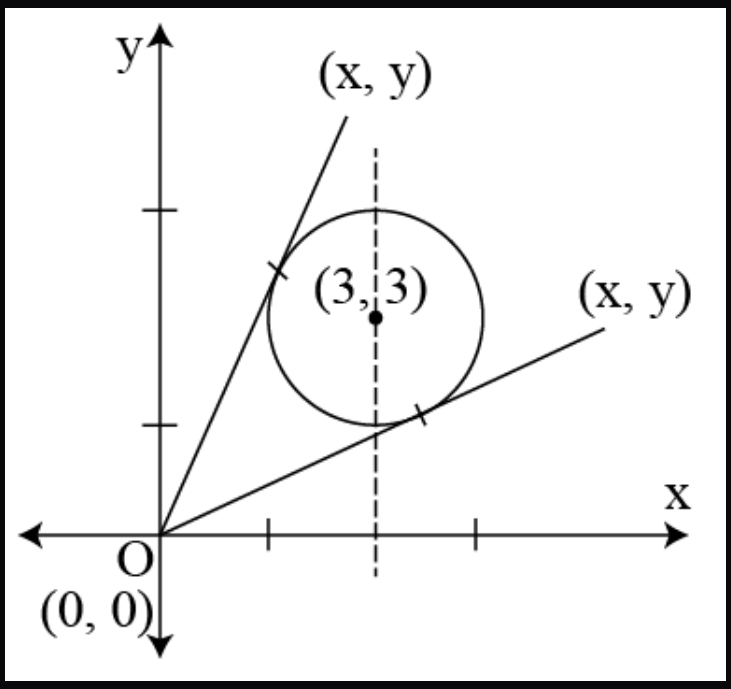
Solution:

$V_1 =$ **variance of** $\{13,16,19 \ldots\ldots\ldots 103\}$
$=$ **variance of** $\{3,6,9 \ldots\ldots\ldots 93\}$
$= 9$ **of variance of** $\{1,2,3 \ldots\ldots\ldots 31\}$
$V_2 =$ **variance of** $\{20,26,32 \ldots\ldots\ldots 200\}$
$=$ **variance of** $\{6,12,18 \ldots\ldots\ldots 186\}$
$= 36$ **of variance of** $\{1,2,3 \ldots\ldots\ldots 31\}$
$\frac{V_1}{V_2} = \frac{1}{4}$

Q.71 Answer:

—

Solution:



Let $\frac{y}{x} = m'$**. So, on putting** $y = m'x$ **in the given circle, We get** $(x-3)^2 + (m'x-3)^2 = 6$
$\Rightarrow x^2\left(1+m^2\right) - 6x\left(1+m'\right) + 12 = 0$
Putting discriminant $= 0$**, we get,** $m' = 3 \pm 2\sqrt{2}$
So, $m = 3 - 2\sqrt{2}$ **and** $M = 3 + 2\sqrt{2} \Rightarrow mM = 1$
$\Rightarrow \frac{1}{M} + \frac{1}{m} = m + M$
$= 3 - 2\sqrt{2} + 3 + 2\sqrt{2}$
$= 6$

Q.72 Answer:

—

Solution:

$z^3 + (\bar{\omega})^7 = 0 \ldots$ **(i)**
$\Rightarrow z^3 = -(\bar{\omega})^7$
$z^{15} = -(\bar{\omega})^{35} \ldots (ii)$

-

and $z^5 \cdot \omega^{11} = 1 \quad \ldots$ (iii)

Taking cube at both sides we get,

$z^{15} \cdot \omega^{33} = 1$

$-(\bar{\omega})^{35} \cdot \omega^{33} = 1 \ldots$ (iv)

(from equation (ii))

Taking modulus at both sides,

$|\overline{(\omega)}^{35} \cdot \omega^{33}| = 1$

$\Rightarrow |\omega|^{68} = 1$

$\Rightarrow |\omega| = 1$

So

$|\omega|^2 = 1$

$\Rightarrow \omega\bar{\omega} = 1$

$\Rightarrow \bar{\omega} = \dfrac{1}{\omega}$

Taking modulus in ((11),

$|z^5 \cdot \omega^{11}| = 1$

$\Rightarrow |z|^5 |\omega|^{11} = 1$

Hence, $|z| = 1$.

Now from equation (iv),

$-\left(\frac{1}{\omega}\right)^{35} \cdot \omega^{33} = 1$

$\Rightarrow \omega^2 = -1 = i^2$

$\Rightarrow \omega = \pm i$

$z = \pm i$

So, 4 ordered pairs of (z, ω) are possible $(i, i), (i. -i), (-i, -i) \& (-i, i)$.

Q.73 Answer:

—

Solution:

We know, area of a triangle

$\Delta = \frac{1}{2} \times$ base $\times$ altitude

$\therefore \frac{\left(\frac{1}{p_1}\right)^2 + \left(\frac{1}{p_3}\right)^2}{\left(\frac{1}{p_2}\right)^2} = \frac{\left(\frac{2\Delta}{p_1}\right)^2 + \left(\frac{2\Delta}{p_3}\right)^2}{\left(\frac{2\Delta}{p_2}\right)^2} = \frac{(BC)^2 + (AB)^2}{(AC)^2}$ {where Δ represents the area of triangle ABC}

$= \frac{(5-3)^2 + (2-(-1))^2 + (-1-5)^2 + (1-2)^2}{(-1-3)^2 + (1-(-1))^2}$

$= \frac{4+9+36+1}{16+4} = \frac{50}{20} = \frac{5}{2}, a = 5$ and $b = 2$

Q.74 Answer:

—

Solution:

There are three cases for this:

(a) Apples are distributed as $(4,\ 3,\ 1)$.

Therefore, total number of ways $= {}^8C_4 \times {}^4C_3 \times 3!$

$= 70 \times 4 \times 6$

$= 1680$

(b) Apples are distributed as $(4,\ 2,\ 2)$.

Therefore, total number of ways $= {}^3C_2 \times {}^8C_4 \times {}^4C_2$

$= 3 \times 70 \times 6$

$= 1260$

(c) Apples are distributed as $(3,\ 3,\ 2)$.

Therefore, total number of ways $= {}^3C_2 \times {}^8C_3 \times {}^5C_3 = 1680$

So, total number of ways including all the cases $= 1680 + 1260 + 1680$

$= 22 \times {}^7P_3$

Therefore, $k = 22$.

Q.75 Answer:

—

Solution:

Let, the last element of set A **is,** $t_{200} = 2 + (200 - 1)2 = 400$

and the last element of set B **is** $t_{250} = 3 + (250 - 1)3 = 750$

Common elements are $\rightarrow 6, 12, 18 \ldots$

We have to find the common elements upto 400

$t_n = 6 + (n - 1)6 \Rightarrow 400 = 6 + 6n - 6 \Rightarrow n = \frac{400}{6} = 66.6 \ldots$

$t_{66} = 6 + (66 - 1)6 = 396$

Number of common elements $= 66$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 200 + 250 - 66 = 384 \Rightarrow k = 384$